

Spectral Theory of Hilbert spaces

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Introduction

“If one had to pick one result that stood out among the many in the theory of operators, especially self-adjoint operators, on a Hilbert space, it would be the spectral theorem ...”

— Barry Simon [Sim15]

For a self-adjoint linear operator (or equivalently a Hermitian matrix) on $\mathbb{C}^n$, there is always a basis consisting of eigenvectors. And the operator has a diagonal matrix representation. The elements on the diagonal are its eigenvalues. In the infinite-dimensional setting, the picture is not as simple as in the finite-dimensional case. In the early 1900, Hilbert’s work on integral equations gave birth to the notion of Hilbert spaces. He developed the spectral theory of bounded self-adjoint operators on a Hilbert space. Many follow-up important developments were mainly due to his students.

However, the development of the theory for unbounded self-adjoint operators happened much later in 1929 by three young mathematicians, John von Neumann, Marshall Stone, and Aurel Wintner independently. The main motivation was to provide a mathematical foundation for developments in quantum mechanics. The work of von Neuman and Stone has lasting impact because they use the language of operators [Sim15].

In quantum mechanics, the state of a quantum particle is described by a vector in a separable Hilbert space. Observable quantities such as position, energy, and momentum are represented by (unbounded) self-adjoint operators. For example, the classical coordinate x_j is represented by the multiplication operator M_{x_j} , and the momentum component p_j by the differential $-i\hbar\frac{d}{dx}$, where $\hbar$ is the Planck constant. The evolution of states is described by the Schrödinger equation. The spectral theorem provides a powerful tool to study the self-adjoint operators. Our goal is to cover basic tools and fundamental theorems in the spectral theory of bounded and unbounded self-adjoint operators in infinite-dimensional Hilbert spaces. Our emphasis from the beginning will be on unbounded operators. We illustrate the theory by discussing important examples such as the Laplacian, Schrödinger operator, etc.

There are many great textbooks on the subject. The main references used for this lecture note are [Bor20, Dav95, Hel13, Sim15].

Chapter 1

Bounded & Unbounded Linear Operators

1.1 Basic definitions

We consider a Hilbert space $\mathcal{H}$ over $\mathbb{C}$ with scalar product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ which is antilinear in its second argument. We assume that $\mathcal{H}$ is a separable Hilbert space.

Examples of a Hilbert spaces that we will consider are $L^2(\mathbb{R}^n)$, $L^2(\Omega)$ and Sobolev spaces $H^1(\Omega)$, $H^2(\Omega)$ where Ω is a domain in $\mathbb{R}^n$ or in a Riemannian manifold. We will give the definition of Sobolev spaces in the next section.

Some of the natural linear operators that we would like to consider are multiplication operator, differentiation d/dx , Laplacian $\Delta = -\sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}$, the harmonic oscillator $-\frac{d^2}{dx^2} + x^2$. You may notice that the domains of these operators cannot be for example the whole of $L^2(\mathbb{R})$ or $L^2(\mathbb{R}^n)$. So, it is natural to consider linear operators on $\mathcal{H}$ whose domains are not equal to $\mathcal{H}$.

Definition 1.1

A **linear operator** L on $\mathcal{H}$ is a linear map defined from a dense subspace $D(L)$ of $\mathcal{H}$ to $\mathcal{H}$. The subspace $\mathcal{D}(L)$ is called the domain of L . Let $\tilde{L}$ be a linear operator on $\mathcal{H}$ with domain $\mathcal{D}(\tilde{L}) \supsetneq \mathcal{D}(L)$ and $L = \tilde{L}$ on $\mathcal{D}(L)$. Then we call $\tilde{L}$ an **extension** of L .

Basically, we need a pair $(L, \mathcal{D}(L))$ to identify the linear operator. Knowing L is not enough to identify its domain $\mathcal{D}(L)$. But when there is no confusion we may use L to represent the pair $(L, \mathcal{D}(L))$.

Example 1.1 (1D-Laplacian). Consider the one-dimensional Laplacian $\Delta f = -f''$ on $L^2([0, 1])$. We consider different choices of $\mathcal{D}(\Delta)$:

- $\mathcal{D}(\Delta) = \{f \in C^2([0, 1]) : -f'(0) + \alpha f(0) = f'(1) + \alpha f(1) = 0\}$, $\alpha > 0$.
- $\mathcal{D}(\Delta) = \{f \in C^2([0, 1]) : f(0) = f(1) = 0\}$. In this case, we denote the pair $(\Delta, \mathcal{D}(\Delta))$ by Δ_D .
- $\mathcal{D}(\Delta) = \{f \in C^2([0, 1]) : f'(0) = f'(1) = 0\}$. In this case, we denote the pair $(\Delta, \mathcal{D}(\Delta))$ by Δ_N .

Note that none of these operators are extension of the other. But all of them are an extension of Δ with domain equal to $C_c^2([0, 1])$.

The following example will play a central role in our discussion even though it does not seem to be related to a differential operator.

Example 1.2 (Multiplication Operator). Let X be a Borel set in $\mathbb{R}$ and μ be a measure which is finite on any bounded Borel set of $\mathbb{R}$. Let $\phi : X \rightarrow \mathbb{C}$ be a measurable function such that it is bounded on any bounded Borel subset of X . The linear map M_ϕ on $L^2(X, \mu) := \{f : X \rightarrow \mathbb{C} : \int_X |f|^2 d\mu < \infty\}$

$$M_\phi f = \phi f$$

with domain $D(M_\phi) := \{f \in L^2(X, \mu) : \phi f \in L^2(X, \mu)\}$ is called the **multiplication operator**. It is a densely defined operator which cannot be extended to the whole $L^2(X, \mu)$ in general. The multiplication operator M_x on $L^2(\mathbb{R})$ is called the **position operator** in Quantum mechanics.

Example 1.3 (Quantum Harmonic Oscillator). Let $H = \Delta + V(x)$ where $V(x)$ is a positive definite quadratic form and $\mathcal{D}(H) = \mathcal{S}(\mathbb{R}^n) = \{f \in C^\infty(\mathbb{R}^n) : \|x^\beta D^\alpha f\|_\infty < \infty\}$ is the Schwartz space of rapidly decaying functions. Here $\alpha = (\alpha_1, \dots, \alpha_n)$, and $\beta = (\beta_1, \dots, \beta_n)$ $\alpha_i, \beta_i \in \mathbb{N}$, and

$$x^\beta = x_1^{\beta_1} \dots x_n^{\beta_n}, \quad D^\alpha f := \frac{\partial^{\alpha_1} \dots \partial^{\alpha_n}}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}} f.$$

If L is continuous on $\mathcal{D}(L)$, it has a unique continuous extension to $\mathcal{H}$. Equivalently, L is bounded, i.e., $\|L\| := \sup_{x \neq 0} \frac{\|L(x)\|}{\|x\|} \leq C$ for some positive constant C . So considering a bounded linear operator with a densely defined domain doesn't give a generalisation. Hence, for a bounded linear operator L , we always assume that $\mathcal{D}(L) = \mathcal{H}$ and denote the space of bounded linear operators on $\mathcal{H}$ by $\mathcal{L}(\mathcal{H})$.

The main object of interest for us is the study of operators that are not necessarily bounded. It gives us a framework to study linear differential operators. Unbounded operators are not continuous on $\mathcal{D}(L)$. However, there is a natural way to get very close to the definition of continuity or boundedness. The **closed graph theorem** states that a linear map L with $\mathcal{D}(L) = \mathcal{H}$ is continuous if and only if its graph $\Gamma(L) := \{(x, Lx) : x \in \mathcal{H}\}$ is closed in $\mathcal{H} \times \mathcal{H}$. When $\mathcal{D}(L) \subsetneq \mathcal{H}$ we can still consider $\Gamma(L)$ and ask if $\Gamma(L)$ is a closed subset of $\mathcal{H} \times \mathcal{H}$. And if it is not, can its closure be considered as the graph of a linear operator?

Definition 1.2

A linear operator L is called **closed** if $\Gamma(L)$ is closed in $\mathcal{H} \times \mathcal{H}$. Or equivalently, L is closed if for every sequence (x_n) in $\mathcal{D}(L)$ such that $x_n \rightarrow x$ and $Lx_n \rightarrow y$, for some $x, y \in \mathcal{H}$ we have $x \in \mathcal{D}(L)$ and $Lx = y$.

We say that L is **closable** if $\overline{\Gamma(L)}$ is the graph of an operator. We denote this operator by $\bar{L}$ and call it the **closure** of L . Notice that $\overline{\Gamma(L)} = \Gamma(\bar{L})$.

But why we are interested in the notion of closedness? To motivate the importance of Definition 1.2, Let us first define/recall some terminologies.

Definition 1.3: Symmetric & Self-Adjoint Operators

Let L be a linear operator on $\mathcal{H}$ with domain $\mathcal{D}(L)$. We say L is **symmetric** (aka Hermitian) if

$$\langle Lx, y \rangle = \langle x, Ly \rangle, \quad \forall x, y \in \mathcal{D}(L).$$

The adjoint operator L^* with domain

$$\mathcal{D}(L^*) = \{y \in \mathcal{H} : \exists z \in \mathcal{H} \text{ such that } \langle Lx, y \rangle = \langle x, z \rangle, \forall x \in \mathcal{D}(L)\}^a$$

is given by the following defining identity

$$\langle Lx, y \rangle = \langle x, L^*y \rangle, \quad \forall x \in \mathcal{D}(L), \quad \forall y \in \mathcal{D}(L^*).$$

And we say L is a **self-adjoint operator** if $L = L^*$ and $\mathcal{D}(L) = \mathcal{D}(L^*)$. The linear operator L is called **essentially self-adjoint** if L is closable and $\bar{L}$ is self-adjoint.

^aNote that z is unique because $\mathcal{D}(L)$ is dense in $\mathcal{H}$.

Example 1.4. Let consider the multiplication operator M_x on $L^2(\mathbb{R})$. It is easy to check that it is a symmetric operator. We show that M_x is a self-adjoint operator. It is enough to verify that $\mathcal{D}(M_x^*) = \mathcal{D}(M_x)$. Let $g \in \mathcal{D}(M_x^*)$. Then by definition, there exist $h \in L^2(\mathbb{R})$ such that $\langle M_x f, g \rangle = \langle f, h \rangle$ for every $f \in \mathcal{D}(M_x)$. By the Cauchy-Schwartz inequality

$$|\langle M_x f, g \rangle|^2 \leq C \|f\|^2, \quad \forall f \in \mathcal{D}(M_x),$$

where $C > 0$ is a constant depending on g . Take $f(x) = xg(x)\chi_{[-R, R]}(x)$ in the above inequality. Here, $\chi_{[-R, R]}(x)$ is the characteristic function of $[-R, R]$. Then, after simplification, we get

$$\int_{\mathbb{R}} x^2 |g(x)|^2 \chi_{[-R, R]}(x) \leq C \quad \forall R > 0.$$

Sending R to $+\infty$, we conclude that $xg(x) \in L^2(\mathbb{R})$, i.e. $g \in \mathcal{D}(M_x)$. The convergence is guaranteed by the dominated convergence theorem.

Note that the adjoint operator is always closed. See Exercise 1.4. However, the domain of L^* is not necessarily dense in $\mathcal{H}$. The density of the domain of L^* is linked to the closeability of L .

Proposition 1.1

The linear operator L with domain $\mathcal{D}(L)$ is closable if and only if $\mathcal{D}(L^*)$ is dense. Moreover, $\bar{L} = L^{**}$

We first prove this following useful lemma.

Lemma 1.1. $\Gamma(L^*) = J(\Gamma(L))^\perp$, where $J(x, y) = (-y, x)$.

Proof.

$$\begin{aligned}
\Gamma(L^*) &= \{(x, L^*x) : x \in \mathcal{D}(L^*)\} \\
&= \{(x, y) : x \in \mathcal{D}(L^*), \langle z, y - L^*x \rangle = 0, \forall z \in \mathcal{D}(L)\} \\
&= \{(x, y) : x \in \mathcal{D}(L^*), \langle z, y \rangle - \langle Lz, x \rangle = 0, \forall z \in \mathcal{D}(L)\} \\
&= \{(x, y) \in \mathcal{H} \times \mathcal{H} : \langle z, y \rangle - \langle Lz, x \rangle = 0, \forall z \in \mathcal{D}(L)\} \\
&= \{(x, y) \in \mathcal{H} \times \mathcal{H} : \langle J(z, Lz), (x, y) \rangle = 0, \forall z \in \mathcal{D}(L)\} \\
&= J(\Gamma(L))^\perp
\end{aligned} \tag{1.1}$$

where identity (1.1) follows from the definition of $D(L^*)$. $\square$

Proof Proposition 1.1. Assume that L is closable. Take $x \in \mathcal{D}(L^*)^\perp$. Then for every $z \in \mathcal{D}(L)$

$$0 = \langle x, z \rangle = \langle (0, x), (-L^*z, z) \rangle = \langle (0, x), J(z, L^*z) \rangle.$$

Hence, $(0, x) \in J(\Gamma(L^*))^\perp = J(J\Gamma(L)^\perp)^\perp = \overline{\Gamma(L)}$. Since L is closable, $x = \bar{L}0 = 0$ and we conclude that $H = \overline{\mathcal{D}(L^*)}$.

Now assume that $\mathcal{D}(L^*)$ is dense. We shall see $\overline{\Gamma(L)}$ can be viewed as the graph of L^{**} which is a closed operator by Lemma 1.1. Let's consider $\Gamma(L^{**})$.

$$\Gamma(L^{**}) = \Gamma((L^*)^*) = J(\Gamma(L^*))^\perp = J(J(\Gamma(L))^\perp)^\perp = \overline{\Gamma(L)}.$$

We conclude that $\bar{L} = L^{**}$. It completes the proof. $\square$

As an immediate corollary we get

Corollary 1.1. Every symmetric operator L is closable.

We can show that the closure of a symmetric operator is also symmetric. See Exercise 1.3. We can easily verify that Δ_D and Δ_N are symmetric and therefore closable. But they are not closed operators (why?).

In the theory of bounded linear operators, there is no difference between the definition of a symmetric operator and self-adjoint operator since the domain of L is $\mathcal{H}$. But it is not the case for an unbounded operator. We can easily see that when L is symmetric, then L^* is an extension of L . For a symmetric linear operator L , there are two natural extensions:

$$\bar{L} \quad \text{and} \quad L^*.$$

Any self-adjoint extension $(\tilde{L}, \mathcal{D}(\tilde{L}))$ of $(L, \mathcal{D}(L))$ (if it exists) is the restriction of L^* to $\mathcal{D}(\tilde{L})$ and an extension of $\bar{L}$. (why?) Of course, when a symmetric operator is essentially self-adjoint, then its closure is the unique self-adjoint extension (why?). The question is when a symmetric operator has a self-adjoint extension and when its self-adjoint extension is unique. We postpone answering this question to Section 1.4.

1.2 Resolvent and Spectrum

We now define the central concepts of this lecture.

Definition 1.4

Let L be a linear operator on $\mathcal{H}$ with domain $\mathcal{D}(L)$. The **resolvent set** $\rho(L)$ of L is

$$\rho(L) := \left\{ z \in \mathbb{C} : \begin{array}{l} (z - L) \text{ is a bijection from } \mathcal{D}(L) \text{ to } \mathcal{H}, \\ \text{and } (z - L)^{-1} \text{ is bounded} \end{array} \right\}.$$

The operator $R(z; L) := (z - L)^{-1}$ is called the **resolvent operator** of L or the resolvent for short. When there is no confusion, we may drop L . The **spectrum** of L is

$$\sigma(L) := \mathbb{C} \setminus \rho(L).$$

A complex number λ is an **eigenvalue** of L if there exists $0 \neq f \in \mathcal{H}$ such that $Lf = \lambda f$. Here, f is called an **eigenfunction** of L . The set of eigenvalues is also called the **point spectrum** and denoted by $\sigma_p(L)$.

Example 1.5. Consider the multiplication operator M_ϕ on $L^2([0, 1])$, where $\phi \in C^0([0, 1])$ and μ is the Lebesgue measure. In this case, one can show that $\sigma(M_\phi) = \phi([0, 1])$ and $\sigma_p(M_\phi) = \{\lambda : \mu(\phi^{-1}(\lambda)) > 0\}$. In particular, if ϕ is monotone, $\sigma_p(M_\phi) = \emptyset$.

Remark 1.1. If $\rho(L)$ is nonempty, then L must be closed; See Exercise 1.6. So if L is not closed then $\sigma(L) = \mathbb{C}$. The reverse statement is not true. There are closed (unbounded) operators whose resolvent set is empty.

Our goal is to classify operators using their spectrum. The interesting situation is when $\sigma(L) \neq \mathbb{C}$, otherwise we cannot say much. Among closed operators, those that are bounded, i.e. elements of $\mathcal{L}(\mathcal{H})$, always have nonempty resolvent set.

Theorem 1.1

- a) For every $L \in \mathcal{L}(\mathcal{H})$, $\sigma(L)$ is a nonempty compact subset of $B(0, \|L\|) \subset \mathbb{C}$.
- b) For every self-adjoint operator $L \in \mathcal{L}(\mathcal{H})$, $\sigma(L) \subseteq [m, M] \subset \mathbb{R}$, where

$$m = \inf_{x \neq 0} \frac{\langle Lx, x \rangle}{\|x\|^2}, \quad M = \sup_{x \neq 0} \frac{\langle Lx, x \rangle}{\|x\|^2}.$$

Moreover, $m, M \in \sigma(L)$.

We leave the proof of part a) as an exercise. See Exercise 1.9. For part b), we use the following important lemma. It can be considered as a generalisation of the Riesz-Fréchet theorem.

Lemma 1.2 (Lax-Milgram Lemma). Let $B : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ be a sesquilinear form, i.e. $B(x, y)$ is linear in x and antilinear in y . Assume that

- i) there exists $c > 0$ such that

$$|B(x, y)| \leq c\|x\|\|y\|, \quad \forall x, y \in \mathcal{H}, \quad (\text{continuity});$$

ii) there exists $b > 0$ such that

$$|B(x, x)| \geq b\|x\|^2, \quad \forall x \in \mathcal{H}, \quad (\text{coercivity}).$$

Then one can define the linear operator $A \in \mathcal{L}(\mathcal{H})$ by $\langle Ax, y \rangle = B(x, y)$. It gives a bijection and $A^{-1} \in \mathcal{L}(\mathcal{H})$.

It is a good exercise to prove this theorem yourself. You can find the proof of the Lax-Milgram lemma for example in [Hel13, Theorem 3.4].

Proof of Theorem 1.1. b) Let $z \in \mathbb{C}$ with $\text{Im } z \neq 0$. Then

$$|\langle (z - L)x, x \rangle| = |(\text{Re } z)\|x\|^2 - \langle Lx, x \rangle + i(\text{Im } z)\|x\|^2| \geq |\text{Im } z|\|x\|^2.$$

Now, applying the Lax-Milgram lemma to the sesquilinear form $B(x, y) := \langle (z - L)x, y \rangle$, we conclude that $z \in \rho(L)$. Therefore $\sigma(L) \subset \mathbb{R}$. If $\lambda > M$ or $\lambda < m$, apply the Lax-Milgram lemma to $B(x, y) = \langle (\lambda - L)x, y \rangle$ or $B(x, y) = \langle (L - \lambda)x, y \rangle$ to conclude that $\lambda \in \rho(L)$. We now show that $M \in \sigma(L)$. Let (x_n) be a sequence in $\mathcal{H}$ with $\|x_n\| = 1$ such that $\langle (M - L)x_n, x_n \rangle \rightarrow 0$ as $n \rightarrow \infty$. We define $\langle x, y \rangle := \langle (M - L)x, y \rangle$ and apply the Cauchy-Schwarz inequality

$$|\langle x, y \rangle| \leq \langle x, x \rangle^{1/2} \langle y, y \rangle^{1/2}$$

to $x = x_n$ and $y = (M - L)x_n$. We get

$$\begin{aligned} \|(M - L)x_n\|^2 &\leq \langle (M - L)x_n, x_n \rangle^{1/2} \langle (M - L)^2 x_n, (M - L)x_n \rangle^{1/2} \\ &\leq \langle (M - L)x_n, x_n \rangle^{1/2} \|M - L\| \|(M - L)x_n\|. \end{aligned}$$

Therefore $\|(M - L)x_n\| \rightarrow 0$. If $M \in \rho(L)$, then $\|(M - L)^{-1}(M - L)x_n\| = \|x_n\| \rightarrow 0$ which is a contradiction. The proof of $m \in \sigma(L)$ is similar. □

Remark 1.2. When L is nonnegative, i.e. $\langle Lx, x \rangle \geq 0$, then $\|L\| = \sup_{\|x\| \neq 0} \frac{\langle Lx, x \rangle}{\|x\|^2} = M$.

When L is closable, it is natural to consider the spectrum and resolvent set of $\bar{L}$ instead of L . We consider self-adjoint operators. We shall see that for this family of operators the spectral theorem gives a nice classification.

Theorem 1.2

Let L be a self-adjoint operator on $\mathcal{H}$. Then $\emptyset \neq \sigma(L) \subset \mathbb{R}$. Moreover, if $z \notin \mathbb{R}$ then

$$\|(z - L)^{-1}\| \leq \frac{1}{|\text{Im } z|}.$$

Exercise 1.8 gives an interesting bound for the norm of the resolvent.

Proof. If $\sigma(L) = \emptyset$, then L^{-1} is a bounded self-adjoint operator and its spectrum is a subset of $\mathbb{R}$. Take any $0 \neq \lambda \in \mathbb{R}$. Then $L^{-1} - \lambda$ has a bounded inverse $\lambda^{-1}L(\lambda^{-1} - L)^{-1}$ (why is it bounded?). Therefore, $\sigma(L^{-1}) = \{0\}$ and by Theorem 1.1 b), L^{-1} must be 0. It is a contradiction and we conclude that $\sigma(L) \neq \emptyset$.

The proof of $\sigma(L) \subset \mathbb{R}$ is very similar the bounded case. We need an extension of Lax-Milgram lemma, where the sesquilinear form B is defined on a dense subspace V of $\mathcal{H}$: Similar to the proof of Theorem 1.1, we have

$$|\langle (z - L)x, x \rangle| \geq |\operatorname{Im} z| \|x\|^2;$$

$$\frac{1}{|\operatorname{Im} z|} \|(z - L)x\| \geq \|x\|. \quad (1.2)$$

It shows that $(z - L)$ is one-to-one. We need to show that its range is equal to $\mathcal{H}$. We first show that it is closed. Let $(z - L)x_n \rightarrow y$. Then by (1.2), (x_n) is Cauchy and it converges to $x \in \mathcal{H}$. Therefore, $Lx_n \rightarrow zx - y$. We now show that $x \in \mathcal{D}(L)$. For any $w \in \mathcal{D}(L)$

$$\langle Lx_n, w \rangle = \langle x_n, Lw \rangle.$$

After taking limit as $n \rightarrow \infty$

$$\langle zx - y, w \rangle = \langle x, Lw \rangle.$$

By the definition of $\mathcal{D}(L^*)$, we conclude that $x \in \mathcal{D}(L^*) = \mathcal{D}(L)$ and $Lx = zx - y$. So, the range of $(z - L)$ is closed. If it is not equal to $\mathcal{H}$, there is a nonzero vector v in its orthogonal complement:

$$\langle (L - z)x, v \rangle = \langle Lx, v \rangle - \langle x, \bar{z}v \rangle = 0, \quad \forall x \in \mathcal{D}(L).$$

The second identity implies that $v \in \mathcal{D}(L^*)$ and $Lv = \bar{z}v$. This is a contradiction with $\langle Lv, v \rangle$ being real. Moreover, inequality (1.2) implies that $\|(z - L)^{-1}\| \leq \frac{1}{|\operatorname{Im} z|}$. $\square$

The following theorem can be considered as the converse of Theorem 1.2.

Theorem 1.3

Let L be a symmetric operator such that $\mathbb{C} \setminus \mathbb{R} \subset \rho(L)$. Then L is self-adjoint.

Proof. Here, we just give the steps of the proof and we leave the details as an exercise.

Take $z, \bar{z} \in \mathbb{C} \setminus \mathbb{R}$ and show that $(L - z)^{-1}$ is the adjoint of $(L - \bar{z})^{-1}$.

By definition $y \in \mathcal{D}(L^*)$ with $L^*y = v$ if for all $x \in \mathcal{D}(L)$

$$\langle Lx, y \rangle = \langle x, v \rangle.$$

We want to show that $y \in \mathcal{D}(L)$. Subtracting $z\langle x, y \rangle$ from both sides of the above identity, we get

$$\langle (L - z)x, y \rangle = \langle x, v - \bar{z}y \rangle.$$

Let $w := (L - z)x$ and rewrite it in terms of w to obtain $y = (L - \bar{z})^{-1}(v - \bar{z}y)$. Notice that $\operatorname{Ran}(L - \bar{z})^{-1} = \mathcal{D}(L)$. With this, conclude the statement. $\square$

We end this section with an important theorem about the spectra of self-adjoint operators. It shows that we can see the spectrum as “approximate eigenvalues”.

Theorem 1.4

Let L be a self-adjoint operator. Then $\lambda \in \sigma(L)$ if and only if there exists a sequence

(x_n) in $\mathcal{D}(L)$ with $\|x_n\| = 1$ and $\|(L - \lambda)x_n\| \rightarrow 0$.

Proof. If $\|(L - \lambda)x_n\| \rightarrow 0$ with $\|x_n\| = 1$ we show that $\lambda \notin \rho(L)$. If $\lambda \in \rho(L)$ then

$$\|x_n\| = \|(\lambda - L)^{-1}y_n\| \leq \|(\lambda - L)^{-1}\| \|y_n\|.$$

Hence,

$$\|(\lambda - L)(x_n)\| \geq \frac{1}{\|(\lambda - L)^{-1}\|}$$

and we get a contradiction. Notice, we didn't use the self-adjointness of L for this part.

If $\lambda \in \sigma_p(L)$ the proof is trivial. Let $\lambda \in \sigma(L) \setminus \sigma_p(L) \subset \mathbb{R}$. Then $L - \lambda$ is injective. If it is surjective then $(L - \lambda)^{-1}$ is a symmetric operator defined on $\mathcal{H}$ and therefore bounded (see Exercise 1.3). Thus, the operator $L - \lambda$ must not be surjective. Since¹ $\{0\} = \ker(L - \lambda) = \text{Ran}(L - \lambda)^\perp$, we conclude that $\text{Ran}(L - \lambda)$ is dense in $\mathcal{H}$ and $(L - \lambda)^{-1} : \text{Ran}(L - \lambda) \rightarrow \mathcal{H}$ is unbounded. Let (y_n) be a sequence in $\text{Ran}(L - \lambda)$ with $\|y_n\| = 1$ such that $\|(L - \lambda)^{-1}y_n\| \rightarrow \infty$. Now define $x_n := \frac{(L - \lambda)^{-1}y_n}{\|(L - \lambda)^{-1}y_n\|}$. It has the desired property. $\square$

1.3 Sobolev Spaces

Let $\Omega \subset \mathbb{R}^n$ be an open and $\phi \in C^k(\Omega)$ and $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$, $\alpha^i \in \mathbb{N}$ be a multi-index with $|\alpha| = \alpha_1 + \dots + \alpha_n = k$. Then we denote

$$D^\alpha \phi := \frac{\partial^{\alpha_1} \dots \partial^{\alpha_n}}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}} \phi$$

Definition 1.5: Weak Partial Derivatives

Let $\Omega \subset \mathbb{R}^n$. We say $g \in L^1_{\text{loc}}(\Omega)$ is the α -weak partial derivative of $f \in L^1_{\text{loc}}(\Omega)$ if for every $\phi \in C_c^\infty(\Omega)$ we have

$$\int_{\Omega} D^\alpha \phi \bar{f} dx = (-1)^{|\alpha|} \int_{\Omega} \phi \bar{g} dx \quad (1.3)$$

Notice that when f is in $C^{|\alpha|}(\Omega)$, by integration by part we get

$$\int_{\Omega} D^\alpha \phi \bar{f} dx = (-1)^{|\alpha|} \int_{\Omega} \phi D^\alpha \bar{f} dx$$

for every $\phi \in C_c^\infty(\Omega)$. Hence, the weak partial derivative g is equal to $D^\alpha f$. The weak derivative, if it exists, is unique up to a set of measure zero.

Remark 1.3. Note that for any $f \in L^1_{\text{loc}}(\Omega)$, we can define the distributional derivative in the following sense.

$$\langle \phi, D^\alpha f \rangle = (-1)^{|\alpha|} \langle D^\alpha \phi, f \rangle, \quad \forall \phi \in C_c^\infty(\Omega).$$

But we call it a weak derivative when it is in $L^1_{\text{loc}}(\Omega)$. By a distribution, we mean a linear functional from $C_c^\infty(\Omega) \rightarrow \mathbb{C}$.

¹It is an easy exercise in functional analysis to show that $\ker(L^*) = \text{Ran}(L)^\perp$.

Definition 1.6: Sobolev Spaces on $\mathbb{R}^n$

Let $\Omega \subset \mathbb{R}^n$ be an open set. For a non-negative integer k and $p \in [1, \infty]$, the **Sobolev space** $W^{k,p}(\Omega)$ is defined as follows.

$$W^{k,p}(\Omega) := \{f : \Omega \rightarrow \mathbb{C} : D^\alpha f \in L^p(\Omega), \forall |\alpha| \leq k\},$$

with natural norm

$$\|f\|_{W^{k,p}} := \left(\sum_{|\alpha| \leq k} \int_{\Omega} |D^\alpha f|^p \right)^{1/p}.$$

A important subspace of $W^{k,p}(\Omega)$ is the closure of $C_c^\infty(\Omega)$ in $W^{k,p}(\Omega)$:

$$W_0^{k,p}(\Omega) := \overline{C_c^\infty(\Omega)}.$$

When $p = 2$, $W^{k,2}(\Omega)$ and $W_0^{k,2}$ are Hilbert spaces and we denote them by $H^k(\Omega)$ and $H_0^k(\Omega)$ respectively.

In the next example, we can see that we can use the Fourier transform to calculate the weak derivative. Let us first recall some basic facts about the Fourier transform²

$$\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$$

$$\mathcal{F}(f)(\xi) = \hat{f}(\xi) := \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx.$$

- For any multi-index α and $f \in \mathcal{S}$ we have

$$\mathcal{F}(D^\alpha f)(\xi) = (i\xi)^\alpha \hat{f}(\xi).$$

- (**Plancherel's theorem**) The Fourier transform $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ is a bijection and preserve the L^2 inner product . It extends uniquely to a unitary map $\mathcal{F} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$ and its inverse is the unique unitary extension of

$$\mathcal{F}^{-1} : \mathcal{S} \rightarrow \mathcal{S}$$

$$\mathcal{F}^{-1}(f)(x) = \check{f}(x) := \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} f(\xi) e^{ix \cdot \xi} d\xi.$$

Example 1.6. Let $f \in L^2(\mathbb{R}^n)$. Then $D^\alpha f \in L^2(\mathbb{R}^n)$ if and only if $\xi^\alpha \hat{f} \in L^2(\mathbb{R}^n)$. This is a consequence of the fact that $\mathcal{F}$ preserves the L^2 inner product. Let us assume that $\xi^\alpha \hat{f} \in L^2(\mathbb{R}^n)$. Define $g := \mathcal{F}^{-1}((i\xi)^\alpha \hat{f}) \in L^2(\mathbb{R})$. Then for every $\phi \in C_c^\infty(\Omega)$ we have

$$\begin{aligned} \langle \mathcal{F}^{-1}((i\xi)^\alpha \hat{f}), \phi \rangle &= \langle (i\xi)^\alpha \hat{f}, \hat{\phi} \rangle \\ &= (-1)^{|\alpha|} \langle \hat{f}, (i\xi)^\alpha \hat{\phi} \rangle \\ &= (-1)^{|\alpha|} \langle \hat{f}, \mathcal{F}(D^\alpha \phi) \rangle \\ &= (-1)^{|\alpha|} \langle f, D^\alpha \phi \rangle \end{aligned}$$

²There are different conventions for the definition of the Fourier transform in the literature. If you are interested to learn about possible conventions and the relation between them you may visit the link below.
<http://www.johndcook.com/blog/fourier-theorems/>.

which shows that $g = D^\alpha f$. The above argument can be easily reversed to show that if $f, D^\alpha f \in L^2(\mathbb{R}^n)$, then $\xi^\alpha \hat{f} \in L^2(\mathbb{R}^n)$.

We can therefore conclude that

$$\begin{aligned} H^k(\mathbb{R}^n) &= \left\{ f \in L^2(\mathbb{R}^n) : \xi^\alpha \hat{f} \in L^2(\mathbb{R}^n), \forall |\alpha| \leq k \right\} \\ &= \left\{ f \in L^2(\mathbb{R}^n) : \langle \xi \rangle^k \hat{f} \in L^2(\mathbb{R}^n) \right\}, \end{aligned}$$

where $\langle \xi \rangle := (1 + |\xi|^2)^{1/2}$.

Example 1.7. Let consider Δ on $L^2(\mathbb{R}^n)$ with $\mathcal{D}(\Delta) = C_c^\infty(\mathbb{R}^n)$. Then for every $\phi \in C_c^\infty(\mathbb{R}^n)$,

$$\mathcal{F}(\Delta\phi)(\xi) = |\xi|^2 \hat{\phi}(\xi) = M_{|\xi|^2} \hat{\phi}(\xi),$$

where $M_{|\xi|^2}$ is the multiplication operator on $L^2(\mathbb{R}^n)$ with $\mathcal{D}(M_{|\xi|^2}) = \{f \in L^2(\mathbb{R}^n) : |\xi|^2 f \in L^2(\mathbb{R}^n)\}$. By abuse of notation $\Delta := \mathcal{F}^{-1} M_{|\xi|^2} \mathcal{F}$ with $\mathcal{D}(\Delta) = H^2(\mathbb{R}^n)$ is an extension of the Laplacian acting on $C_c^\infty(\Omega)$, and it is unitarily equivalent to the multiplication operator $M_{|\xi|^2}$. Moreover, it is a self-adjoint extension. See Exercise 1.1. In fact, you can show that it is essentially self-adjoint (Exercise 1.14).

1.4 Self-Adjoint Extensions of Symmetric Operators

In this section we address the following questions: give a symmetric operator L

- Does it have a self-adjoint extension?
- When is its self-adjoint extension (if it exists) unique?
- Can we construct its self-adjoint extension (if it exists)?

To illustrate the problem of self-adjoint extensions of a symmetric operator, we give some examples.

Let us first state the following useful proposition. We leave its proof as an exercise (see Exercise 1.3).

Proposition 1.2

Let L be a symmetric operator on $\mathcal{H}$ with domain $\mathcal{D}(L)$. For any $z \in \mathbb{C} \setminus \mathbb{R}$, $z - \bar{L}$ maps $\mathcal{D}(\bar{L})$ one-to-one onto a closed subspace of $\mathcal{H}$.

Example 1.8. Let $L = -i \frac{d}{dx}$ on $L^2(\mathbb{R})$ with $\mathcal{D}(L) = C_c^1(\mathbb{R})$.

Claim. It is symmetric and essentially self-adjoint.

One way to show the above claim is to follow the same lines of argument as in Example 1.7 using the Fourier transform. Here, we present a different approach which will be useful in the following examples. It is easy to verify that L is symmetric using integration by part. Let z be any complex number in $\mathbb{C} \setminus \mathbb{R}$.

$$\text{Ran}(z - L) = \left\{ f \in L^2(\mathbb{R}) : zu + i \frac{du}{dx} = f, \text{ for some } u \in C_c^1(\mathbb{R}) \right\}.$$

Multiplying $zu + i \frac{du}{dx} = f$ by e^{-izx} we get

$$i \frac{d}{dx} (e^{-izx} u) = e^{-izx} f.$$

Since $u \in C_c^1(\mathbb{R})$, we should have

$$\int_{\mathbb{R}} e^{-izx} f = 0. \quad (1.4)$$

Conversely, every continuous compactly supported f satisfying (1.4) is in $\text{Ran}(z - L)$. Indeed, for $f \in C_c(\mathbb{R})$, take $u(x) := -i \int_{-\infty}^x e^{-iz(y-x)} f(y) dy \in C_c^1(\mathbb{R})$. Since $e^{-izx} \notin L^2(\mathbb{R})$, the set of continuous functions with compact support satisfying (1.4) is dense in $L^2(\mathbb{R})$ (why?). According to Proposition 1.2, the range of $z - \bar{L}$ is closed. Hence, $\text{Ran}(z - \bar{L}) = L^2(\mathbb{R})$. Since $(z - \bar{L})$ is one-to-one, $z \in \rho(\bar{L})$. According to Theorem 1.3, $\bar{L}$ is self-adjoint.

Example 1.9. Let $L = -i \frac{d}{dx}$ on $L^2(\mathbb{R}_+)$ with $\mathcal{D}(L) = C_c^1(\mathbb{R}_+)$.

Claim. It is symmetric but it has no self-adjoint extension.

Symmetry of L of course follow from the integration by part. Following a similar argument as in the previous example, every $f \in C_c(\mathbb{R}_+)$ satisfying

$$\int_0^\infty e^{-izx} f = 0 \quad (1.5)$$

is in $\text{Ran}(z - L)$. For $\text{Im } z > 0$, e^{-izx} is not in $L^2(\mathbb{R}_+)$. Therefore, using the same argument as before $\text{Ran}(z - \bar{L}) = L^2(\mathbb{R}_+)$ for any $z \in \mathbb{C}$ with $\text{Im } z > 0$. If $\text{Im } z < 0$, $e^{-izx} \in L^2(\mathbb{R}_+)$ and $\text{Ran}(z - \bar{L}) = \{f \in L^2(\mathbb{R}_+) : f \perp e^{-izx}\} \neq L^2(\mathbb{R}_+)$. According to Theorem 1.2, $\bar{L}$ is not self-adjoint.

To show that it has no self-adjoint extension, we note that any self-adjoint extension of L should be an extension of $\bar{L}$. Let $\tilde{L}$ be a self-adjoint extension. Then according to Theorem 1.2, $\mathbb{C} \setminus \mathbb{R} \subset \rho(\tilde{L})$. Take z with $\text{Im } z > 0$. We know $\text{Ran}(z - \bar{L}) = L^2(\mathbb{R}_+)$. Let $y \in \mathcal{D}(\tilde{L}) \setminus \mathcal{D}(\bar{L})$. Then there exists $x \in \mathcal{D}(\bar{L})$ such that

$$(z - \bar{L})x = (z - \tilde{L})y.$$

Therefore, $(z - \tilde{L})(x - y) = 0$. It implies that $x = y \in \mathcal{D}(\bar{L})$ and gives a contradiction.

Example 1.10. Let $L = -i \frac{d}{dx}$ on $L^2(0, 1)$ with $\mathcal{D}(L) = C_c^1(0, 1)$.

Claim. It is symmetric but not essentially self-adjoint. However, it has infinitely many self-adjoint extensions.

Again, the symmetry of L follows by integration by part. Arguing as before, for any z with $\text{Im } z \neq 0$ we have $\text{Ran}(z - \bar{L}) = \{f \in L^2(0, 1) : f \perp e^{-izx}\} \neq L^2(0, 1)$. Therefore, $\bar{L}$ is not self-adjoint according to Theorem 1.2.

To see that it has infinitely many self-adjoint extensions, take any $1 \neq \alpha \in \mathbb{C}$ with $|\alpha| = 1$ and consider $L_\alpha = -i \frac{d}{dx}$ with $\mathcal{D}(L_\alpha) = \{u \in C^1(0, 1) : u(1) = \alpha u(0)\}$. Then $\bar{L}_\alpha$ is a self-adjoint extension of L . See Exercise 15.

For a given symmetric operator L , we have

$$\|(L + i)x\|^2 = \|Lx\|^2 + \|x\|^2 = \|(L - i)x\|^2.$$

Therefore, $C = (L - i)(L + i)^{-1}$ defines an isometric linear mapping from $\text{Ran}(L + i)$ one-to-one onto $\text{Ran}(L - i)$. This map is called the **Cayley transform**.

Lemma 1.3. There exists a one-to-one correspondence between symmetric extension of L and isometric extension of its Cayley transform.

We skip the proof of this lemma. You can find the proof in [Dav95, Lemma 1.2.6].

Definition 1.7

The **deficiency subspaces** of a symmetric operator L are

$$D^\pm := \ker(L^* \pm i) = \text{Ran}(L \mp i)^\perp$$

and $n_\pm := \dim D^\pm$ is called the **deficiency index**.

Remark 1.4. a) The codimension of $\text{Ran}(L - z)$ for all z with $\text{Im } z > 0$ is the same. Similarly the codimension of $\text{Ran}(L - z)$ for all z with $\text{Im } z < 0$ is the same, see for example [Lax02, Chapter 33].

b) The deficiency indices of L and $\bar{L}$ are the same.

We are now ready to state the best-known condition for essential self-adjointness.

Theorem 1.5

Let L be a symmetric operator on $\mathcal{H}$ with domain $\mathcal{D}(L)$. Then L has a self-adjoint extension if and only if $n_+ = n_-$. Moreover, the following statements are equivalent

- i) L is essentially self-adjoint.
- ii) $n_\pm = 0$.

Proof. i) $\Rightarrow$ ii) If $x \in \bar{D}^+ = \text{Ran}(\bar{L} - i)^\perp$, then $i\langle x, x \rangle = \langle \bar{L}x, x \rangle$. But this contradicts with the fact that $\langle \bar{L}x, x \rangle$ is real. So, $n_+ = 0$. Similarly we get $n_- = 0$.

ii) $\Rightarrow$ i) According to Proposition 1.2 $(\bar{L} + i)$ maps $\mathcal{D}(\bar{L})$ one-to-one onto a closed subspace of $\mathcal{H}$. Since $n_- = 0$, $\text{Ran}(\bar{L} + i) = \mathcal{H}$. Therefore, for any $x \in \mathcal{D}(L^*)$, there exists $y \in \mathcal{D}(\bar{L})$ such that

$$(\bar{L} + i)y = (L^* + i)x.$$

Notice that L^* is an extension of $\bar{L}$. Hence, $(L^* + i)(x - y) = 0$. Therefore, $x - y \in \ker(L^* + i)$. Since $n_+ = 0$, we get $x = y \in \mathcal{D}(L^*)$. Therefore, $\bar{L} = L^*$.

We now prove the first statement of the theorem. If $n_+ = n_- \neq 0$, then there exists an extension of the Cayley transform to $\mathcal{H}$ as a unitary map. By Lemma 1.3, this corresponds to a symmetric extension $\tilde{L}$ of L . The deficiency indices of $\tilde{L}$ is 0, since its Cayley transform is unitary. So, $\tilde{L}$ is a self-adjoint extension of L . Note that when the deficiency indices are equal but not zero, there are infinitely many self-adjoint extensions by Lemma 1.3.

We now prove the reverse. Let $\tilde{L}$ be a self-adjoint extension of L . Then the deficiency indices of $\tilde{L}$ must be zero because of [i) $\Rightarrow$ ii)]. The Cayley transform of $\tilde{L}$ is an extension of the Cayley transform of L to $\mathcal{H}$. It gives a bijection between D^+ and D^- . Hence, $n^+ = n^-$. $\square$

Corollary 1.2. Let L be a closed symmetric operator on $\mathcal{H}$. L is essentially self-adjoint if and only if $L^* = \bar{L}$.

Proof. Exercise 1.16 $\square$

It in particular, it implies that L has exactly one self-adjoint extension if it is essentially self-adjoint.

When self-adjoint extensions of a symmetric operator are not unique, we would like that the extension enjoys some of the same properties of the original operator e.g. positivity and semi-boundedness.

Definition 1.8

Let L be a symmetric operator on $\mathcal{H}$. We say that L is **semi-bounded** (from below) if there exists $C \in \mathbb{R}$ such that

$$\langle Lx, x \rangle \geq C\|x\|^2, \quad \forall x \in \mathcal{D}(L).$$

When C can be chosen equal to 0, we say L is **positive**.

Theorem 1.6: Friedrichs Extension

A symmetric semibounded operator L on $\mathcal{H}$ with domain $\mathcal{D}(L)$ admits a self-adjoint extension $\tilde{L}$ such that $\tilde{L}$ is also semibounded satisfying the same bound.

Proof. Up to adding a constant we assume that

$$\langle Lx, x \rangle \geq \|x\|^2 \quad \forall x \in \mathcal{D}(L). \quad (1.6)$$

Let us define the sesquilinear form

$$(x, y)_L := \langle x, Ly \rangle.$$

It defines a new inner product on $\mathcal{D}(L)$. We consider the completion $\mathcal{H}_L$ of $\mathcal{D}(L)$ with respect to L -norm which consists of equivalent classes of Cauchy sequences. Note that (1.6)

$$\|x\| \leq \|x\|_L$$

implies that every Cauchy sequence in L -norm is Cauchy in the norm of $\mathcal{H}$ and has a limit in $\mathcal{H}$. It gives a natural map from $\mathcal{H}_L$ to $\mathcal{H}$. The question is if this natural map is injective. In other words, can we consider the completion $\mathcal{H}_L$ of $\mathcal{D}(L)$ in L -norm as a subspace of $\mathcal{H}$?

Lemma 1.4. . The natural mapping of $\mathcal{H}_L$ into $\mathcal{H}$ gives an embedding, i.e. it is one-to-one.

Proof. See Exercise 1.17. □

So, we can view $\mathcal{H}_L$ as a dense subspace of $\mathcal{H}$. Now, we define the extension L_F of L by

$$\langle y, L_F x \rangle = (y, x)_L, \quad \forall y \in \mathcal{H}_L, x \in \mathcal{D}(L_F)$$

with domain

$$\mathcal{D}(L_F) := \{x \in \mathcal{H}_L : |(y, x)_L| \leq c\|y\| \text{ for all } y \in \mathcal{H}_L\}.$$

Note that we use the Riesz-Frechet Theorem on $\mathcal{H}_L$ to uniquely define $L_F x$. Indeed, for any $z \in \mathcal{H}$, $\langle \cdot, z \rangle$ is continuous on $\mathcal{H}_L$ thanks to (1.6). $(\mathcal{H}_L, (\cdot, \cdot)_L)$ is a Hilbert space. Hence, by Riesz-Frechet theorem, there exist $x \in \mathcal{H}_L$ such that $\langle y, z \rangle = (y, x)_L$. In particular, it shows that L_F maps $\mathcal{D}(L_F)$ one-to-one onto $\mathcal{H}$. Since $(\cdot, \cdot)_L$ and $\langle \cdot, \cdot \rangle$ are sesquilinear, then L_F is symmetric and linear. It is easy to check that it is an extension of L . It remains to show that

it is self-adjoint. According to Exercises 1.3 and 1.11, L_F^{-1} is symmetric and bounded. Notice that $\mathcal{D}(L_F^{-1}) = \mathcal{H}$ and therefore, L_F^{-1} is self-adjoint. By Theorem 1.1, $\mathbb{C} \setminus \mathbb{R} \subset \rho(L_F^{-1})$. Take any $z \in \mathbb{C} \setminus \mathbb{R}$. The identity

$$z - L_F = L_F(L_F^{-1} - z^{-1})z$$

shows that $\mathbb{C} \setminus \mathbb{R} \subset \rho(L_F)$. Because the right-hand side of the identity is invertible and its inverse is bounded. Now, Theorem 1.3 implies that L_F is self-adjoint. $\square$

Although, the Friedrichs extension theorem provides a self-adjoint extension for any semi-bounded operator, it doesn't determine whether it is essentially self-adjoint. From the proof of Theorem 1.5, we can get the following criteria for essentially self-adjointness of positive operators. We leave its proof as an exercise. Of course, any semi-bounded operator after adding a constant can be considered a positive operator.

Corollary 1.3. Let L be a positive symmetric operator on $\mathcal{H}$. Then the following are equivalent.

- i) L is essentially self-adjoint.
- ii) $\ker(L^* + 1) = 0$.
- iii) $\text{Ran}(L + 1)^\perp = 0$.

1.5 Examples

In this section we study the self-adjoint extension of some of the most important operators in quantum mechanics. We postpone the discussion on their spectra to the next chapter.

Dirichlet Laplacian

It describes to a number of physical problems including the heat flow in a medium with boundary remaining at zero temperature at all times, vibrations of an elastic membrane with fixed boundary, and more relevant to us, the motion of a quantum particle confined to a region by the barrier associated with very large potential.

Let Ω be a bounded open set in $\mathbb{R}^n$. Consider Δ on $L^2(\Omega)$ with $\mathcal{D}(\Delta) = C_c^\infty(\Omega)$. It is a positive symmetric operator but it is not essentially self-adjoint (why? you may use the comment after Example 1.1). We now study its Friedrichs extension. We consider $L := \Delta + 1$ in order to have

$$\langle f, Lf \rangle = \langle \nabla f, \nabla f \rangle + \|f\|^2 \geq \|f\|^2, \quad \forall f \in C_c^\infty(\Omega).$$

It is easy to see that the completion of $C_c^\infty(\Omega)$ in $L^2(\Omega)$ with respect to $(\cdot, \cdot)_L$ is $H_0^1(\Omega)$. As we discuss in the previous section, we can define a self-adjoint operator L_F by

$$\langle g, L_F f \rangle = (g, f)_L, \quad \forall g \in H_0^1(\Omega), f \in \mathcal{D}(L_F)$$

with

$$\begin{aligned} \mathcal{D}(L_F) &= \{f \in H_0^1(\Omega) : |(g, f)_L| \leq c\|g\|, \forall g \in H_0^1(\Omega)\} \\ &= \{f \in H_0^1(\Omega) : \Delta f \in L^2(\Omega)\}. \end{aligned}$$

When Ω is regular, we can give a better description of $\mathcal{D}(L_F)$ thanks to some elliptic regularity.

Elliptic Regularity. Let Ω be an open set in $\mathbb{R}^n$ such that $\bar{\Omega}$ is compact. Assume that $\partial\Omega$ is C^2 smooth. If $f \in H_0^1(\Omega)$ and $\Delta f \in L^2(\Omega)$, then $f \in H^2(\Omega)$.

So, in that case $\mathcal{D}(L_F) = H_0^1(\Omega) \cap H^2(\Omega)$. The self-adjoint operator $L_F - 1$ is the Friedrichs extension of Δ and we call it the Dirichlet Laplacian Δ_D . And its domain is $\mathcal{D}(\Delta_D) = \{f \in H_0^1(\Omega) : \Delta f \in L^2(\Omega)\}$.

As an exercise, consider Δ with domain $\{f \in C^\infty(\bar{\Omega}) : \partial_n f = 0\}$, where $\partial_n f$ is the normal derivative of f along the boundary. Can you identify the domain of its Friedrichs extension?

We now give a couple of examples for which we can calculate the spectrum of the Dirichlet Laplacian. In both examples, the spectrum consists of eigenvalues. In the next chapter, we will see that the spectrum of the Dirichlet Laplacian when the domain Ω is bounded is equal to its point spectrum.

Example 1.11. Let $\Omega = (0, 1)^n \subset \mathbb{R}^n$. Then the functions

$$f_k(x) := \prod_{i=1}^n \sin(k_i \pi x_i),$$

$k = (k_1, \dots, k_n)$ are a complete set of eigenfunctions of Δ_D with corresponding eigenvalues $\lambda_k := \pi^2(k_1^2 + \dots + k_n^2)$.

Example 1.12. Let $B(0, R)$ is the ball of radius R centred at 0 in $\mathbb{R}^2$. As an exercise you can show that in polar coordinates (r, θ) , eigenfunctions of Δ_D on $B(0, R)$ are of the form

$$J_n(j_{n,s} r/R) e^{in\theta}$$

where $s, n \in \mathbb{N}$ and $j_{n,s}$ is the s th zero of the Bessel functions J_n . The corresponding eigenvalues are $j_{n,s}^2/R^2$.

According to Exercise 1.13, we can conclude that $\sigma(\Delta_D)$ consists of eigenvalues in both examples above (why?).

Schrödinger Operator

Let consider $\Delta + V(x)$ on $C_c^\infty(\mathbb{R}^n)$, where V is a continuous real valued function.

Bounded Potentials. In Example 1.7, we showed that Δ is essentially self-adjoint. According to the following lemma, when V is bounded, $\Delta + V(x)$ is essentially self-adjoint and the domain of its closure is $H^2(\mathbb{R}^n)$.

Lemma 1.5. If L is essentially self-adjoint and T is a bounded self-adjoint operator, then $L + T$ is essentially self-adjoint on $\mathcal{D}(L)$.

Proof. Clearly $\bar{L} + T$ is symmetric on $\mathcal{D}(\bar{L})$. It easy to verify that $\overline{\bar{L} + T} = \bar{L} + T =: A$. Let $\mathcal{D}(A) := \mathcal{D}(\bar{L})$ and $x \in \mathcal{D}(A^*)$. Then there exists $z \in \mathcal{H}$ such that

$$\langle (\bar{L} + T)y, x \rangle = \langle y, z \rangle, \quad \forall y \in \mathcal{D}(L).$$

Since $\langle Ty, x \rangle = \langle y, Tx \rangle$ for every $x, y \in \mathcal{H}$, we get

$$\langle \bar{L}y, x \rangle = \langle y, z - Tx \rangle, \quad \forall y \in \mathcal{D}(L).$$

Therefore, $x \in \mathcal{D}(\bar{L})$ and A is self-adjoint. □

Semi-Bounded Potentials. What if V is not bounded? Many important examples of Schrödinger operators don't have a bounded potential e.g. the quantum Harmonic oscillator $\Delta + |x|^2$. In this case, the potential is positive. In general, if there exists $c \in \mathbb{R}$ such that $V(x) > c$, then $\Delta + V(x)$ is semibounded. For simplicity, we assume that $V(x) \geq 0$. By the Friedrichs extension we know that it has a self-adjoint extension. Let us identify its domain. Let assume $V \in L_{\text{loc}}^\infty(\mathbb{R}^n)$. We consider $L := \Delta + V + 1$ and the associated sesquilinear form

$$(f, g)_L := \langle \nabla f, \nabla g \rangle + \langle V^{1/2} f, V^{1/2} g \rangle + \langle f, g \rangle, \quad f, g \in C_c^\infty(\mathbb{R}^n).$$

The completion of $C_c^\infty(\mathbb{R}^n)$ in $L^2(\mathbb{R}^n)$ w.r.t. $(\cdot, \cdot)_L$ is

$$D_L = \{f \in H^1(\mathbb{R}^n) : V^{1/2} f \in L^2(\mathbb{R}^n)\}.$$

Then we get

$$\begin{aligned} \mathcal{D}(L_F) &:= \{f \in D_L : |(f, g)_L| \leq c \|g\|, \forall g \in D_L\} \\ &= \{f \in D_L : (\Delta + V + 1)f \in L^2(\mathbb{R}^n)\} \subset H_{\text{loc}}^2(\mathbb{R}^n). \end{aligned}$$

The last inclusion is because $V \in L_{\text{loc}}^\infty(\mathbb{R}^n)$.

However, it doesn't tell if it is essentially self-adjoint.

Theorem 1.7

Let $V \in L_{\text{loc}}^\infty(\mathbb{R}^n)$ be a real valued function with $V \geq 0$. Then $\Delta + V$ with domain $C_c^\infty(\mathbb{R}^n)$ is essentially self-adjoint.

Proof. Let $A = \Delta + V$. According to Corollary 1.3, it is enough to show that $A^* + 1$ is injective. If $f \in \mathcal{D}(A^*)$, then by definition $g \mapsto \langle (\Delta + V)g, f \rangle$ is bounded on $C_c^\infty(\mathbb{R}^n)$. It is equivalent to say that $(\Delta + V)f \in L^2(\mathbb{R}^n)$. Since $V \in L_{\text{loc}}^\infty(\mathbb{R}^n)$, we get $f \in H_{\text{loc}}^2(\mathbb{R}^n)$. Therefore, $\mathcal{D}(A^*) \subset H_{\text{loc}}^2(\mathbb{R}^n)$. Let $(A^* + 1)f = 0$. It means that for every $\phi \in C_c^\infty(\mathbb{R}^n)$

$$\langle f, (\Delta + V + 1)\phi \rangle = 0.$$

It is continuous in ϕ w.r.t. H^2 -norm. Let $\tau : \mathbb{R} \rightarrow \mathbb{R}$ be a smooth function so that it is 1 for $t < 0$ and 0 for $t > 1$. Let $\rho_r(x) = \tau(|x| - r)$, $r > 0$. Since $f \in H_{\text{loc}}^2(\mathbb{R}^n)$, there is a sequence in $C_c^\infty(\mathbb{R}^n)$ approximating $\rho_r f$ in H^2 -norm. Passing to the limit, we obtain that for any $r > 0$

$$\langle f, (\Delta + V + 1)\rho_r f \rangle = 0.$$

We now conclude by letting $r \rightarrow \infty$. □

Example 1.13 (Quantum Harmonic Oscillator). Consider the 1-dimensional Harmonic oscillator

$$H := -\frac{d^2}{dx^2} + x^2.$$

We can calculate its eigenvalues and eigenfunctions explicitly. We just cite the result and refer the details of calculation to [Bor20, Section 7.1.4]. The eigenfunctions and corresponding eigenvalues are of the form

$$f_k(x) = H_k(x)e^{-x^2/2}, \quad \lambda_k = 2k + 1$$

where H_k is the k -th Hermite polynomial. The Hermite polynomials form an orthogonal basis for $L^2(\mathbb{R}, e^{-x^2/2}dx)$. Hence, by Exercise 1.13,

$$\sigma(H) = 2\mathbb{N} + 1.$$

The harmonic oscillator in higher dimensions can be reduced to solving the 1-dimensional case. In $\mathbb{R}^n$, up to a change of basis, any harmonic oscillator can be written in the form of $\Delta + \sum_{j=1}^n a_j^2 x_j^2$. In this case, the eigenfunctions and respective eigenvalues are given by

$$f_{k_1, \dots, k_n}(x) = \prod_{j=1}^n H_{k_j}(\sqrt{|a_j|}x_j) e^{-|a_j|x_j^2/2}, \quad \lambda_{k_1, \dots, k_n} = \sum_{j=1}^n |a_j|(2k_j + 1).$$

Coulomb potential. An important Schrödinger operator which is not covered by the above cases is the Hamiltonian operator for the Hydrogen atom. It is given by $\Delta - \frac{1}{|x|}$ on $\mathbb{R}^3$. We state a theorem due to Kato without proof which covers the case of the Coulomb potential. You can find more details in [Bor20, Chapter 7].

Theorem 1.8

Let $n \leq 3$, and V be a real-valued potential in $L^2(\mathbb{R}^n) + L^\infty(\mathbb{R}^n)$. Then $\Delta + V$ with domain $C_c^\infty(\mathbb{R}^n)$ is essentially self-adjoint.

The Coulomb potential $\frac{1}{|x|} \in L_{\text{loc}}^2(\mathbb{R}^3)$ and it is in $L^\infty(\mathbb{R}^3)$ outside a compact set. Therefore, according to the above theorem, $\Delta - \frac{1}{|x|}$ is essentially self-adjoint.

The above theorem however doesn't give any information about the domain of the operator. Even though the Coulomb potential is not semi-bounded, $\Delta - \frac{1}{|x|}$ with domain $C_c^\infty(\mathbb{R})$ is semi-bounded. This is the consequence of an important inequality called the Hardy inequality.

Hardy inequality. For any $f \in C_c^\infty(\mathbb{R}^3)$ we have

$$\int_{\mathbb{R}^3} |x|^{-2} |f(x)|^2 dx \leq 4 \int_{\mathbb{R}^3} |\xi|^2 |\hat{f}(\xi)|^2 d\xi.$$

We can rewrite the Hardy inequality as follows

$$\int_{\mathbb{R}^3} |x|^{-2} |f(x)|^2 dx \leq 4 \langle \Delta f, f \rangle.$$

Hence,

$$\int_{\mathbb{R}^3} \frac{1}{|x|} f^2(x) dx \leq \left(\int_{\mathbb{R}^3} |x|^{-2} |f(x)|^2 dx \right)^{1/2} \|f\| \leq 2 \langle \Delta f, f \rangle^{1/2} \|f\|.$$

Therefore, for any $\epsilon > 0$,

$$\int_{\mathbb{R}^3} \frac{1}{|x|} f^2(x) dx \leq \epsilon \langle \Delta f, f \rangle + \frac{1}{\epsilon} \|f\|^2.$$

Taking $\epsilon = 1$, we get

$$\left\langle \left(\Delta - \frac{1}{|x|} \right) f, f \right\rangle \geq -\|f\|^2.$$

We take $\Delta - \frac{1}{|x|} + 2$ to meet the assumption in the proof of the Friedrichs extension theorem. We leave it as an exercise to deduce that the domain of its closure is $H^2(\mathbb{R}^3)$.

1.6 Exercises

- 1.1 a) Show that $D(M_\phi)$ in Example 9 is dense in $L^2(X, \mu)$.
 b) Show that M_ϕ is self-adjoint if and only if ϕ is real valued.
- 1.2 a) For a linear operator L on $\mathcal{H}$ with domain $\mathcal{D}(L)$ show that $\ker(L^*) = \text{Ran}(L)^\perp$.
 b) Show that for a bounded self-adjoint operator L we have $\|L\| = \sup_{\|x\|=1} \langle Lx, x \rangle$.
Hint: you may use the identity $\text{Re}(Lx, y) = \frac{1}{4} (\langle L(x+y), x+y \rangle - \langle L(x-y), x-y \rangle)$.
- 1.3 Let L be a symmetric operator on $\mathcal{H}$ with domain $\mathcal{D}(L)$.
 a) Show that the closure of L is symmetric.
 b) For any $z \in \mathbb{C} \setminus \mathbb{R}$, show that $z - \bar{L}$ maps $\mathcal{D}(\bar{L})$ one-to-one onto a closed subspace of $\mathcal{H}$. *Hint.* You may use the idea in the proof of Theorem 1.2.
 c) Show the inverse of a symmetric operator (if it is invertible) is also symmetric.
- 1.4 Show directly (without using Lemma 1.1) that for any linear operator L on $\mathcal{H}$, its adjoint L^* is closed.
- 1.5 Show that L is closable if and only if it has a closed extension $\tilde{L}$. Moreover, $\mathcal{D}(\bar{L}) \subset \mathcal{D}(\tilde{L})$ where $\tilde{L}$ is any closed extension of L .
- 1.6 Assume that $\sigma(L)$ is not equal to the whole of the complex plane $\mathbb{C}$. Then L must be closed.
- 1.7 Show that the resolvent operator satisfies the following identities.
 a) $R(z; L) - R(w; L) = -(z - w)R(z; L)R(w; L)$;
 b) $R(z; L)R(w; L) = R(w; L)R(z; L)$;
 c) $\frac{d}{dz}R(z; L) = -R(z; L)^2$.
- 1.8 Assume that $\rho(L)$ is nonempty. Show that for any $z_0 \in \rho(L)$, $R(\cdot; L)$ is analytic at z_0 and the radius of convergence is $|z - z_0| < \|R(z_0; L)\|^{-1}$. Conclude that $\|R(z; L)\| \geq \frac{1}{\text{dist}(z, \sigma(L))}$.
Hint. You may use the identity $(z - L)R(w; L) = 1 - (w - z)R(w; L)$.
- 1.9 The aim of this exercise is to prove Theorem 1.1 a)
 a) Show that $\sigma(L) \subset \{z \in \mathbb{C} : |z| \leq \|L\|\}$.
 b) Show that $\rho(L)$ is open.
 c) For any given $x, y \in \mathcal{H}$, show that $f(z) = \langle R(z; L)x, y \rangle$, $z \in \rho(L)$, is analytic in z and $f(z) \rightarrow 0$ as $|z| \rightarrow \infty$.
 d) Show that $\rho(L) \neq \mathbb{C}$.

Hint: You may use the fact that for $A \in \mathcal{L}(\mathcal{H})$ with $\|A\| < 1$, $I - A$ is invertible and its inverse $(I - A)^{-1} = \sum_{n=0}^{\infty} A^n$ is in $\mathcal{L}(\mathcal{H})$.

- 1.10 If A is symmetric and closed, show that A is self-adjoint if and only if A^* is symmetric.
- 1.11 If A is a symmetric operator with $\mathcal{D}(A) = \mathcal{H}$, show that A is bounded.
- 1.12 a) Show that $f : [-1, 1] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} -1 & x \in [-1, 0] \\ 1 & x \in (0, 1] \end{cases}$$

is not in $H^1([0, 1])$.

b) If $f \in H^1(\mathbb{R})$, show that $|f| \in H^1(\mathbb{R})$.

- 1.13 Let L be a symmetric operator on $\mathcal{H}$ with domain $\mathcal{D}(L)$. Assume that $\{e_n\}_{n=1}^\infty \subset \mathcal{D}(L)$ be a complete orthonormal sequence in $\mathcal{H}$ such that for every $n \in \mathbb{N}$,

$$Le_n = \lambda_n e_n, \quad \lambda_n \in \mathbb{R}.$$

Consider

$$\tilde{L}x := \sum_{n=1}^{\infty} x_n \lambda_n e_n, \quad x \in \mathcal{D}(\tilde{L}) := \{x \in \mathcal{H} : \sum_n (1 + \lambda_n^2) |x_n|^2 < \infty\},$$

where $x_n := \langle x, e_n \rangle$.

- a) Show that $\sigma(\tilde{L})$ is the closure of $\{\lambda_n\}$ in $\mathbb{R}$.
- b) Show that $\tilde{L}$ is self-adjoint.
- c) Show that $\tilde{L} = \bar{L}$.
- d) Show that the operators in Example 1.1 are essentially self-adjoint.
- 1.14 Show that the operator in Example 1.7 is essentially self-adjoint.
- 1.15 Given any $1 \neq \alpha \in \mathbb{C}$ with $|\alpha| = 1$, let $L_\alpha = -i \frac{d}{dx}$ with $\mathcal{D}(L_\alpha) = \{f \in C^1(0, 1) : u(1) = \alpha u(0)\}$. Show that L_α is essentially self-adjoint.
- 1.16 Prove Corollary 1.2.
- 1.17 Prove Lemma 1.4.
- 1.18 Prove the Hardy inequality.

$$\int_{\mathbb{R}^3} |x|^{-2} |f(x)|^2 dx \leq 4 \int_{\mathbb{R}^3} |\xi|^2 |\hat{f}(\xi)|^2 d\xi.$$

Hint: For any $\alpha \in \mathbb{R}$ and $f \in C_c^\infty(\mathbb{R}^3; \mathbb{R})$, expand the following trivial inequality

$$\int_{\mathbb{R}^3} \left| \nabla f + \alpha \frac{x f(x)}{|x|^2} \right|^2 dx \geq 0.$$

Chapter 2

Spectral Theorem

Most students of mathematics learn quite early and most mathematicians remember till quite late that every Hermitian matrix (and, in particular, every real symmetric matrix) may be put into diagonal form ...

The spectral theorem is widely and correctly regarded as the generalization of this assertion to operators on Hilbert space. It is unfortunate therefore that even the bare statement of the spectral theorem is widely regarded as somewhat mysterious and deep, and probably inaccessible to the nonspecialist ... Another reason the spectral theorem is thought to be hard is that its proof is hard.

—Halmos [Hal63]

2.1 Spectral Theorem for Compact Operators

Let us first recall the spectral theorem in finite dimension.

For a self-adjoint operator A on an n -dimensional inner product space V there exists an orthonormal basis $\{e_j\}_{j=1}^n$ of eigenvectors with corresponding eigenvalues $\{\lambda_j\}_{j=1}^n \subset \mathbb{R}$ such that $Ae_j = \lambda_j e_j$. An equivalent version is that for all self-adjoint matrices (= Hermitian matrices) $A \in M_n(\mathbb{C})$ there is a unitary matrix U such that UAU^{-1} is diagonal.

The spectral theorem for self-adjoint compact operators¹ on an infinite dimensional Hilbert space² $\mathcal{H}$ is very similar to the classification of Hermitian matrices we discussed above.

Theorem 2.1: Hilbert-Schmidt Theorem

Let $L : \mathcal{H} \rightarrow \mathcal{H}$ be a compact self-adjoint operator. Then there exists a complete orthonormal sequence $(e_n)_{n \geq 1}$ consisting of eigenvectors of L with eigenvalues $\lambda_n \in \mathbb{R}$ such that

$$Lx = \sum_{n=1}^{\infty} \lambda_n \langle x, e_n \rangle e_n, \quad x \in \mathcal{H}.$$

Moreover, any nonzero λ_j has finite multiplicities and the only accumulation point of $\{\lambda_j\}$ is 0.

Proof. See Exercise 2.1 for the steps of the proof. □

¹Recall that $L : \mathcal{H} \rightarrow \mathcal{H}$ is compact if the the image any bounded sequence under L has a convergent subsequence in $\mathcal{H}$.

²We assume $\mathcal{H}$ is infinite dimensional and use V to denote a finite dimensional Hilbert space.

For compact operators (not necessarily self-adjoint), we can also state a spectral theorem. It is known as the Riesz-Schauder theorem. Since our focus is on self-adjoint operators, we just state it without a proof.

Riesz-Schauder Theorem. For a compact operator L on $\mathcal{H}$, all elements of $\sigma(L) \setminus \{0\}$ are eigenvalues of finite multiplicity, and 0 is the only possible accumulation point of $\sigma(L)$.

Remark 2.1. Note that for a compact operator L , 0 is always in the spectrum. But it may not be an eigenvalue. See Exercise 2.2.

We cannot expect that for a general self-adjoint operator, the spectrum consists of eigenvalues. We may have $\sigma_p(L) = \emptyset$. See Example 1.5. Nevertheless, it is very interesting to know when the spectrum of an unbounded self-adjoint operator consists of eigenvalues. In the previous chapter, we have seen few examples for which the eigenvalues and eigenfunctions can be explicitly calculated, and then we used Exercise 1.13 to conclude that the spectrum consists of eigenvalues. In general, the standard way is to verify the compactness of its resolvent. Then by the Hilbert-Schmidt Theorem, a self-adjoint compact resolvent admits an orthonormal basis of eigenvectors.

Corollary 2.1. Let L be an unbounded positive self-adjoint operator on $\mathcal{H}$. Then the following are equivalent.

- i) The resolvent $(L + 1)^{-1}$ is compact.
- ii) There exists a complete orthonormal sequence of eigenvectors $\{\phi_n\}$ with corresponding eigenvalues $\lambda_n \geq 0$ such that $\lambda_n \nearrow +\infty$.

Proof. See Exercise 2.4. □

But how can we verify that the resolvent operator is compact?

Theorem 2.2: Rellich Theorem

For a bounded open set $\Omega \subset \mathbb{R}^n$, the inclusion $H_0^1(\Omega) \hookrightarrow L^2(\Omega)$ is compact. Moreover, if we assume $\partial\Omega$ to be Lipschitz^a then the inclusion $H^1(\Omega) \hookrightarrow L^2(\Omega)$ is compact.

^aWe say $\partial\Omega$ is Lipschitz if $\partial\Omega$ can be locally viewed as the graph of a Lipschitz function.

An immediate consequence of the Rellich theorem is the compactness of the resolvent of the Dirichlet and Neumann Laplacian. For example in the case of the Dirichlet Laplacian, $(\Delta_D + 1)^{-1}$ as a map from $L^2(\Omega) \rightarrow L^2(\Omega)$ is compact:

$$L^2(\Omega) \xrightarrow{(\Delta_D + 1)^{-1}} H_0^1(\Omega) \hookrightarrow L^2(\Omega).$$

Therefore, there is a complete orthonormal sequence of $L^2(\Omega)$ consisting of real-valued eigenfunctions. The corresponding eigenvalues form an increasing sequence of non-negative real numbers tending to infinity. And each eigenvalue has finite multiplicity.

The Neumann Laplacian Δ_N is the Friedrichs extension of Δ with domain $\{f \in C^\infty(\bar{\Omega}) :$

$\partial_n f = 0\}$, where $\partial_n f$ is the normal derivative of f along the boundary. Note that here I assume that $\partial\Omega$ is smooth but the regularity assumption can be weakened. You can show that the range of $(\Delta_N + 1)^{-1}$ is in $H^1(\Omega)$. So you can repeat the same argument as in the Dirichlet case.

Remark 2.2. The boundedness of Ω is essential in the statement of the Rellich theorem. See Exercise 2.5.

We cannot explicitly calculate eigenvalues except in a very few examples. However, one can get quantitative estimates for the eigenvalues using the variational characteristic for eigenvalues.

Variational Formula for eigenvalues. Let L be a self-adjoint operator on $\mathcal{H}$. We define

$$\lambda_k := \inf_{V_k} \sup_{0 \neq f \in V_k} \frac{\langle Lf, f \rangle}{\|f\|^2} \quad (2.1)$$

where the infimum is taken over all k -dimensional subspaces V_k of $\mathcal{D}(L)$. The quotient $\frac{\langle Lf, f \rangle}{\|f\|^2}$ is called the Rayleigh quotient.

Remark 2.3. When L is a positive self-adjoint semi-bounded operator with compact resolvent, then V_k can be taken as a k -dimensional subspace of $\mathcal{H}_{L+1}$ (see the definition of $\mathcal{H}_{L+1}$ on page 19). For example, for the Dirichlet Laplacian $\mathcal{H}_{\Delta+1} = H_0^1(\Omega)$ and for the Neumann Laplacian $\mathcal{H}_{\Delta+1} = H^1(\Omega)$.

The following theorem is one of the most important tool for obtaining quantitative estimates of eigenvalues and for comparing eigenvalues of different operators.

Theorem 2.3: Min-Max Principle

Let L be a positive self-adjoint operator with compact resolvent. Then the sequence (λ_n) defined in (2.1) coincides with the eigenvalues of L written in an increasing order counting the multiplicities and $\lambda_n \nearrow \infty$.

Proof. Exercise 2.6. □

We have seen that the spectral theorem of positive self-adjoint operators with compact resolvent is very similar to the example of Hermitian matrices. But what if the resolvent is not compact? What would be the statement of the spectral theorem for a general self-adjoint operator? The beauty of the theory is that the general forms of the spectral theorem are still closely related to the theory of Hermitian matrices. To see the relation we need to restate the spectral theorem for Hermitian matrices in different equivalent forms.

Functional Calculus. Let $\{\sigma_j\}_{j=1}^k$ be the set of distinct eigenvalues of A . Let E_j be the orthogonal projection onto $\{x : Ax = \sigma_j x\}$. Then the spectral theorem is equivalent to saying that A can be written as follows

$$A = \sum_{j=1}^k \sigma_j E_j.$$

Now, for any continuous function $f : \mathbb{R} \rightarrow \mathbb{C}$ we can define

$$f(A) = \sum_{j=1}^k f(\sigma_j) E_j.$$

This is the extension of the definition of $p(A)$, when p is a polynomial.

We will see that for a self-adjoint operator we can also define $f(A)$. Moreover we can derive other formats from the functional calculus version of the spectral theorem.

Multiplication Operator. Define the measure space $(\sigma(A) \times \mathbb{N}, \mu)$, where

$$\mu = \sum_{j=1}^n \delta_{(\lambda_j, j)}.$$

We consider the multiplication operator M_ϕ with $\phi(\lambda, k) = \lambda$ on $L^2(\sigma(A) \times \mathbb{N}, \mu)$. Then A is unitary equivalent to M_ϕ .

$$UAU^{-1}f = M_\phi f$$

with

$$\begin{aligned} U &: \mathbb{C}^n \rightarrow L^2(\sigma(A), \mu) \\ x &\mapsto f_x, \quad f_x(\lambda_j, k) = \langle x, e_j \rangle \delta_{jk}. \end{aligned}$$

We'll see that we get the above version of spectral theorem for self-adjoint operators using the functional calculus form of the spectral theorem.

Projection Valued Measure. Let $\sigma_1, \dots, \sigma_k, k \leq n$ be the distinct eigenvalues of A , and E_j be the orthogonal projection onto $\{v : Av = \sigma_j v\}$, then the spectral theorem is equivalent to

$$A = \sum_{j=1}^k \sigma_j E_j.$$

Equivalently, if we write the sum as integration over the following projection-valued measure

$$dE_t = \sum_{j=1}^k \delta(t - \sigma_j) E_j$$

we get

$$A = \int t dE_t.$$

This form also gives another version of the spectral theorem for a self-adjoint operator.

2.2 Spectral Theorem for Self-adjoint operators

2.2.1 Continuous Functional Calculus

The first version of the spectral theorem we state is the functional calculus form of the spectral theorem. Let $C_0(\mathbb{R})$ be the space of all continuous functions on $\mathbb{R}$ which vanish at $\pm\infty$.

Theorem 2.4: Spectral Theorem (Functional Calculus)

Let L be a given self-adjoint operator. There exists a unique linear map $f \mapsto f(L)$ from $C_0(\mathbb{R})$ to $\mathcal{L}(\mathcal{H})$ satisfying the following properties for all $f \in C_0(\mathbb{R})$.

1. The map $f \mapsto f(L)$ is multiplicative, i.e. $(fg)(L) = f(L)g(L)$;
2. $\bar{f}(L) = f(L)^*$;
3. $\|f(L)\| \leq \|f\|_\infty$;

4. If $w \notin \mathbb{R}$ and $r_w(x) := (w - x)^{-1}$, then $r_w(L) = (w - L)^{-1}$;
5. If $\text{supp}(f) \cap \sigma(L) = \emptyset$ we have $f(L) = 0$.

There are different approaches to prove this version of spectral theorem. The idea of the proof that we discuss here is presented in Davies' book [Dav95]. The proof is based on giving an explicit formula for $f(L)$ as an integral over the resolvent. This approach is due to Helffer and Sjöstrand and the formula for $f(L)$ is known as the Helffer-Sjöstrand formula.

To state the formula, we first define an **almost analytic extension** of a smooth function $f : \mathbb{R} \rightarrow \mathbb{C}$. The extension $\tilde{f} : \mathbb{C} \rightarrow \mathbb{C}$ is defined by

$$\tilde{f}(z) := \left(\sum_{k=0}^n \frac{f^{(k)}(x)(iy)^k}{k!} \right) \tau(y/\langle x \rangle), \quad (2.2)$$

where $\tau(s)$ is a smooth function defined on $\mathbb{R}$ which is equal to 1 for $|s| < 1$ and equal to 0 for $|s| > 2$. The function $\tilde{f}$ is C^∞ in both variables x and y but is not analytic. However, one can show that $|\frac{\partial \tilde{f}(z)}{\partial \bar{z}}| = O(|y|^n)$ as $y \rightarrow 0$, where $\frac{\partial}{\partial \bar{z}} = \frac{1}{2}(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y})$. In particular $\frac{\partial \tilde{f}}{\partial \bar{z}} = 0$ for every $z \in \mathbb{R}$. This is why it is called an almost analytic extension of f .

Second, we need to define the functional space

$$\mathcal{A} = \bigcup_{\beta < 0} \left\{ f : C^\infty(\mathbb{R}) : |f^{(n)}(x)| \leq c_n \langle x \rangle^{\beta-n} \quad \forall n \in \mathbb{N} \right\}.$$

For every $n \geq 1$, we define the norm

$$\|f\|_n := \sum_{k=0}^n \int_{-\infty}^{\infty} |f^{(k)}(x)| \langle x \rangle^{k-1} dx, \quad f \in \mathcal{A}.$$

Definition 2.1: Helffer-Sjöstrand Formula

Let L be a self-adjoint operator. Then for every $f \in \mathcal{A}$ we define

$$f(L) := -\frac{1}{\pi} \int_{\mathbb{C}} \frac{\partial \tilde{f}}{\partial \bar{z}} (z - L)^{-1} dx dy$$

where $\tilde{f}$ is the almost harmonic extension of f defined in (2.2). It can be shown that the exact choice of n and τ is not important. For example we can take $n = 1$.

Let us first make sense of this formula. In Exercise 1.8, we learned that the resolvent is (norm) analytic function of z on $\rho(L)$. Hence, the integrand is norm continuous for $z \in \mathbb{C} \setminus \mathbb{R}$. Hence we can define the integral over any compact subset of $\mathbb{C} \setminus \mathbb{R}$. We can view it as the norm limit of its approximating Riemann sums. We want to have the convergence (in norm) when the compact set tends to the whole of $\mathbb{C} \setminus \mathbb{R}$. The upper bound that we proved in Theorem 1.2 for the norm of the resolvent plays an important role.

In definition (2.2), let us take $n = 1$ and $\sigma(x, y) := \tau(\frac{y}{\langle x \rangle})$. Then

$$\frac{\partial \tilde{f}}{\partial \bar{z}} = \frac{1}{2} i y f'' \sigma + \frac{1}{2} (f + i y f') (\sigma_x + i \sigma_y).$$

The support of partial derivatives of σ is the set

$$A := \{(x, y) : \langle x \rangle < |y| < 2\langle x \rangle\}$$

and the support of σ is

$$B := \{(x, y) : 0 < |y| < 2\langle x \rangle\}.$$

Then an straightforward calculation and using the estimate $\|(z - L)^{-1}\| \leq \frac{1}{|y|}$ show that

$$|\sigma_x + i\sigma_y| \leq c\langle x \rangle^{-1}\chi_A$$

and

$$\left| \frac{1}{2}iyf''\sigma + \frac{1}{2}(f + iyf')(\sigma_x + i\sigma_y) \right| \|(z - L)^{-1}\| \leq c|f''|\chi_B + |f'|\langle x \rangle^{-1}\chi_A + c|f|\langle x \rangle^{-2}\chi_A$$

for some positive constant $c > 0$. Integrating with respect to y and then x , we conclude that

$$\|f(L)\| \leq c\|f\|_2,$$

and in particular, the integral is convergent in norm.

We now show that the map $f \mapsto f(L)$ from $\mathcal{A}$ to $\mathcal{L}(\mathcal{H})$ has all of the properties listed in Theorem 2.4. From that, the proof of Theorem 2.4 immediately follows.

Lemma 2.1. Let L be a given self-adjoint operator. The map $f \mapsto f(L)$ from $\mathcal{A}$ to $\mathcal{L}(\mathcal{H})$ defined above satisfies the following properties for all $f \in \mathcal{A}$.

1. The map $f \mapsto f(L)$ is multiplicative ;
2. $\bar{f}(L) = f(L)^*$;
3. $\|f(L)\| \leq \|f\|_\infty$;
4. If $w \notin \mathbb{R}$ and $r_w(x) := (w - x)^{-1}$, then $r_w(L) = (w - L)^{-1}$;
5. for $f \in C_c^\infty(\mathbb{R})$ with $\text{supp}(f) \cap \sigma(L) = \emptyset$ we have $f(L) = 0$.

Our goal is not to provide a detailed proof of this lemma. We only highlight the idea of the proof. For a detailed discussion we refer to [Dav95, Chapter 2]. The two key useful facts in the proof of the lemma are the following.

- $C_c^\infty(\mathbb{R})$ is dense in $\mathcal{A}$ in the norm $\|\cdot\|_n$ for each $n \geq 1$.
- The operator $f(L)$ is independent of the choices of σ and n .

Idea of the proof of Lemma 2.1. 1. We can use the density $C_c^\infty(\mathbb{R})$ and the resolvent identities in Exercise 1.7 to show that $(fg)(L) = f(L)g(L)$.

2. This is an immediate consequence of the definition of $f(L)$ and the identity $(\bar{z} - L)^{-1} = ((z - L)^{-1})^*$
3. For $f \in \mathcal{A}$, we define $g = c - (c^2 - |f|^2)^{1/2}$, where c can be any constant strictly bigger than $\|f\|_\infty$. One can show that $g \in \mathcal{A}$ and satisfy the following identity

$$f\bar{f} - 2cg + g^2 = 0.$$

The statement then follows by rewriting this identity in terms of $f(L)$ and $g(L)$.

4. This is the key part of the lemma. It shows that the map $f \mapsto f(L)$ is nontrivial. The proof of this part is the hardest part of the proof of the lemma. The idea goes as follows. For $m > 0$ large enough, we consider

$$\Omega_m := \{(x, y) : |x| < m, m^{-1}\langle x \rangle < |y| < 2m\}.$$

By the Stokes theorem

$$\begin{aligned} r_w(L) &= -\frac{1}{\pi} \lim_{m \rightarrow \infty} \int_{\Omega_m} \frac{\partial \tilde{r}_w}{\partial \bar{z}} (z - L)^{-1} dx dy \\ &= -\frac{i}{2\pi} \lim_{m \rightarrow \infty} \int_{\partial \Omega_m} \tilde{r}_w(z) (z - L)^{-1} dz. \end{aligned}$$

The technical step is to show that

$$\lim_{m \rightarrow \infty} \left\| \int_{\partial \Omega_m} (r_w(z) - \tilde{r}_w(z)) (z - L)^{-1} dz \right\| = 0.$$

Then we conclude that

$$r_w(L) = -\frac{i}{2\pi} \lim_{m \rightarrow \infty} \int_{\partial \Omega_m} r_w(z) (z - L)^{-1} dz = (w - L)^{-1}.$$

For the last identity we used the Cauchy theorem and the Residue theorem.

5. If $f \in C_c^\infty(\mathbb{R})$, we can take a finite number of closed curves $\{\gamma_n\}_{n=1}^\infty$ enclosing a region U which contains the support of f and disjoint from $\sigma(L)$. Then the statement follows by applying the Stokes theorem. □

Proof of Theorem 2.4. The existence of $f \mapsto f(L)$ for $f \in \mathcal{A}$ is a consequence of Lemma 2.1. By Stone-Weierstrass Theorem, $\mathcal{A}$ is dense in $C_0(\mathbb{R})$. We use part 3 of Lemma 2.1 for the existence of such extension. To show that the extension has all required properties is rather straight forward. It remains to prove the uniqueness. Assume that there are two maps satisfying properties 1-5. By part 4, they should be equal on the linear subspace

$$\left\{ \sum_{j=1}^k c_j r_{z_j} : c_j \in \mathbb{C}, z_j \in \mathbb{C} \setminus \mathbb{R}, k \in \mathbb{N} \right\}.$$

Again by the Stone-Weierstrass theorem, we can show that this subspace is dense in $C_0(\mathbb{R})$. Now, by part 3, we conclude that the two maps should coincide on $C_0(\mathbb{R})$. □

When we have a sequence of bounded operators, we can talk about convergence in norm and $\mathcal{L}(\mathcal{H})$ with operator norm is Banach. The following definition of convergence for unbounded self-adjoint operators can be considered very close to the convergence in norm for bounded operators.

Definition 2.2: Norm Resolvent Convergence

Let $\{L_n\}$ be a sequence of (bounded or unbounded) self-adjoint operators. We say that L_n converges to a self-adjoint operator L in **norm resolvent** sense if

$$\|(L_n + i)^{-1} - (L + i)^{-1}\| \rightarrow 0.$$

When L_n and L are bounded self-adjoint operators, then the convergence in norm resolvent is equivalent to the convergence in operator norm. See Exercise 2.??.

Remark 2.4. The particular choice of i in the above definition is not important and can be replaced by any $z \in \mathbb{C} \setminus \mathbb{R}$. More precisely, one can show that

$$\|(z - L_n)^{-1} - (z - L)^{-1}\| \leq \frac{9\langle z \rangle^2}{|\operatorname{Im} z|^2} \|(L_n + i)^{-1} - (L + i)^{-1}\|. \quad (2.3)$$

See [Dav95, Lemma 2.6.1] for a proof of this inequality.

Theorem 2.5

Let L_n and L be self-adjoint operators and $L_n \rightarrow L$ in norm resolvent sense. Then for every $f \in C_0(\mathbb{R})$

$$\|f(L_n) - f(L)\| \rightarrow 0.$$

Proof. If $f \in \mathcal{A}$, the Helffer-Sjöstrand formula and (2.3) imply that

$$\begin{aligned} \|f(L_n) - f(L)\| &\leq \frac{1}{\pi} \int_{\mathbb{C}} \left| \frac{\partial \tilde{f}}{\partial \bar{z}} \right| \|(z - L_n)^{-1} - (z - L)^{-1}\| dx dy \\ &\leq \frac{9}{\pi} \|(L_n + i)^{-1} - (L + i)^{-1}\| \int_{\mathbb{C}} \left| \frac{\partial \tilde{f}}{\partial \bar{z}} \right| \frac{\langle z \rangle^2}{|\operatorname{Im} z|^2} dx dy. \end{aligned}$$

We can show that the integral is finite if we choose a suitable almost analytic extension of f . The statement then follows by the density of $\mathcal{A}$ in $C_0(\mathbb{R})$ and Theorem 2.4. $\square$

The convergence in norm-resolvent has the following important consequence.

Corollary 2.2. Let L_n be a sequence of self-adjoint operators converging to a self-adjoint operator L in norm resolvent sense. Then the spectrum of L_n converges to the spectrum of L in the following sense. If $\lambda \notin \sigma(L)$, then there exists $N \in \mathbb{N}$ such that $\lambda \notin \sigma(L_n)$ for all $n \geq N$, and if $\lambda \in \sigma(L)$ then there exists a sequence $\lambda_n \in \sigma(L_n)$ such that $\lim_{n \rightarrow \infty} \lambda_n = \lambda$.

Proof. The proof is a consequence of Theorem 2.4. We leave it as an exercise. See Exercise 2.16. $\square$

2.2.2 Multiplication Operator Form

We now use the functional calculus form to show that any self-adjoint operator is unitary equivalent to a multiplication operator on some L^2 space. This version of the spectral theorem says that every self-adjoint operator is unitary equivalent to a multiplication operator. More precisely,

Theorem 2.6: Spectral Theorem (Multiplication Operator)

Let L be a self-adjoint operator on $\mathcal{H}$. Then there exists a finite measure μ on $\sigma(L) \times \mathbb{N}$ and a unitary operator

$$U : \mathcal{H} \rightarrow L^2(\sigma(L) \times \mathbb{N}, d\mu)$$

such that

$$ULU^{-1}\phi = h\phi, \quad \forall \phi \in U\mathcal{D}(L),$$

where $h : \sigma(L) \times \mathbb{N} \rightarrow \mathbb{R}$, $h(x, n) = x$. And

$$\mathcal{D}(L) = \{f \in \mathcal{H} : hU(f) \in L^2(\sigma(L) \times \mathbb{N}, d\mu)\}.$$

Remark 2.5. In the statement of the theorem above, we can replace $\sigma(L)$ by $\mathbb{R}$. Then $\sigma(L)$ is equal to the *essential range* of h . See Exercise 2.9 for the definition of the essential range.

Definition 2.3: Invariant and Cyclic Subspaces

Let L be a self-adjoint operator on $\mathcal{H}$. Let S be a closed linear subspace of $\mathcal{H}$. We say that S is invariant under L if $(z - L)^{-1}S \subset S$ for all $z \in \mathbb{C} \setminus \mathbb{R}$.

If there exists $v \in \mathcal{H}$ such that $S = \overline{\text{span}}\{(z - L)^{-1}v : z \in \mathbb{C} \setminus \mathbb{R}\}$, we say S is a **cyclic subspace** for L with **cyclic vector** $v \in \mathcal{H}$.

Remark 2.6. When L is a bounded self-adjoint operator then the above definition is equivalent to:

We say S is invariant under L if $LS \subset S$ and S is cyclic for L if for some $v \in \mathcal{H}$ $S = \overline{\text{span}}\{p(L)v : p \text{ is a polynomial}\}$.

Lemma 2.2. For any self-adjoint operator L on $\mathcal{H}$ there exists a sequence of orthogonal cyclic subspaces S_n of $\mathcal{H}$ with cyclic vectors v_n such that $\mathcal{H}$ is equal to the closure of $\bigoplus_{n=1}^{\infty} S_n$.

Proof. Since $\mathcal{H}$ is separable, take a countable dense subspace $\{f_n\}_{n=1}^{\infty}$ of $\mathcal{H}$. Let S_1 be the cyclic subspace generated by $v_1 := f_1$. By induction let us assume that the orthogonal cyclic subspaces $S_1, \dots, S_n$ is given. Let n_k be the smallest index for which $f_{n_k} \notin S_1 \oplus \dots \oplus S_n$. We take v_{n+1} to be the projection of f_{n_k} on the orthogonal complement of $S_1 \oplus \dots \oplus S_n$. If it is zero, then $\mathcal{H} = S_1 \oplus \dots \oplus S_n$. Otherwise we continue the process. If we can continue it without end then $\mathcal{H} = \overline{\bigoplus_{n=1}^{\infty} S_n}$. $\square$

Lemma 2.3. If S is an invariant subspace of a self-adjoint operator L then $S^{\perp}$ is also an invariant subspace for L . Moreover $f(L)S \subset S$ for any $f \in C_0(\mathbb{R})$.

Proof. Exercise 2.14. $\square$

Let us denote the restriction of L to S_n by L_n . The Helffer-Sjöstrand formula and this lemma show that we can reduce all spectral questions concerning L to questions about L_n . The only disadvantage is that the decomposition of $\mathcal{H}$ into cyclic subspaces for L is not canonical.

The proof of Theorem 2.6 now follows from the following special case of the theorem.

Proposition 2.1

Let L be a self-adjoint operator on $\mathcal{H}$. Assume that $\mathcal{H}$ is cyclic with cyclic vector v . Then there exists a finite measure μ on $\sigma(L)$ and a unitary operator

$$U : \mathcal{H} \rightarrow L^2(\sigma(L), d\mu)$$

such that

$$ULLU^{-1}\phi = h\phi, \quad \forall \phi \in U\mathcal{D}(L),$$

where $h : \sigma(L) \rightarrow \mathbb{R}$, $h(x) = x$, and

$$\mathcal{D}(L) = \{f \in \mathcal{H} : hU(f) \in L^2(\sigma(L), d\mu)\}.$$

Sketch of the proof of Proposition 2.1. - Consider the bounded linear functional

$$\ell : C_0(\mathbb{R}) \rightarrow \mathbb{C}$$

$$f \mapsto \langle f(L)v, v \rangle.$$

One can show that $\|\ell\| = \|v\|^2$.

- Note that for $f \geq 0$, by Theorem 2.4, we have

$$\langle f(L)v, v \rangle = \langle f^{1/2}(L)v, f^{1/2}(L)^*v \rangle = \|f^{1/2}(L)v\|^2 \geq 0.$$

- Now we can apply the Riesz-Markov representation theorem. This is the key ingredient of the proof.

Riesz-Markov Theorem. Let X be a locally compact Hausdorff space. For any positive bounded linear functional ℓ on $C_0(X)$, there is a unique regular countably additive Borel measure μ on X such that

$$\ell(f) = \int_X f(x) d\mu(x), \quad \forall f \in C_0(X).$$

The norm of ℓ as a linear functional is the total mass of μ , that is

$$\|\ell\| = \mu(X).$$

And we conclude that there exists a unique finite countably additive measure μ on $\mathbb{R}$ such that

$$\ell(f) = \langle f(L)v, v \rangle = \int_{\mathbb{R}} f(x) d\mu(x) \quad \forall f \in C_0(\mathbb{R}).$$

- The support of μ is subset of $\sigma(L)$ (why?).

- Consider the map

$$T : C_0(\mathbb{R}) \rightarrow L^2(\mathbb{R}, \mu)$$

$$f \mapsto f$$

and observe that

$$\int_{\mathbb{R}} f \bar{g} d\mu = \ell(f \bar{g}) = \langle f(L)v, g(L)v \rangle.$$

- Define the map

$$U : \{f(L)v : f \in C_0(\mathbb{R})\} \rightarrow L^2(\mathbb{R}, d\mu)$$

$$U(f(L)v) = f$$

and extend it as a unitary map from $\mathcal{H}$ to $L^2(\mathbb{R}, d\mu)$. It is possible because $\mathcal{H}$ is cyclic.

- It remains to show that $ULU^{-1}\phi = h\phi$ for all $\phi \in U\mathcal{D}(L)$ and $\mathcal{D}(L) = \{f \in \mathcal{H} : hU(f) \in L^2(\sigma(L), d\mu)\}$. We refer the details of the last step to [Dav95].

□

Proof of Theorem 2.6. According to Lemma 2.2, we can write $\mathcal{H}$ as the closure of cyclic subspaces S_n for L with cyclic vectors v_n . By Proposition 2.1, there exist measures μ_n and unitary operators $U_n : S_n \rightarrow L^2(\sigma(L), d\mu_n)$ such that the restriction L_n of L to S_n is unitary equivalent to multiplication by x on $L^2(\sigma(L), d\mu_n)$. The theorem now follows by defining the measure μ on $\sigma(L) \times \mathbb{N}$ so that its restriction on $\sigma(L) \times \{n\}$ is μ_n . We can choose $\|v_n\|$ such that μ is a finite measure. $\square$

2.2.3 Borel Functional Calculus

The next theorem extends the functional calculus form of the spectral theorem by replacing $C_0(\mathbb{R})$ with the space of bounded Borel measurable functions on $\mathbb{R}$ denoted by $\mathcal{B}$. We say a sequence (f_n) in $\mathcal{B}$ converges monotonically to $f \in \mathcal{B}$ if $\operatorname{Re} f_n(x)$ and $\operatorname{Im} f_n(x)$ converge monotonically to $\operatorname{Re} f(x)$ and $\operatorname{Im} f(x)$ respectively. Note that we do not identify two functions in $\mathcal{B}$ if they are equal everywhere w.r.t the Lebesgue measure.

In the statement we use the notion of the strong convergence of a sequence of bounded operators. Recall that a sequence $T_n \in \mathcal{L}(\mathcal{H})$ converges strongly to $T \in \mathcal{L}(\mathcal{H})$ and we denote by $T_n \xrightarrow{s} T$ if $\lim_{n \rightarrow \infty} \|T_n(x) - T(x)\| = 0$ for all $x \in \mathcal{H}$.

Theorem 2.7: Spectral Theorem (Borel Functional Calculus)

Let L be a self-adjoint operator on $\mathcal{H}$. There exists a map $f \mapsto f(L)$ from $\mathcal{B}$ to $\mathcal{L}(\mathcal{H})$ which is an extension of the map of Theorem 2.4. Moreover, the extension is unique if in addition to properties 1-5 we require

6. $f_n(L) \xrightarrow{s} f(L)$ whenever $f_n \in \mathcal{B}$ converges monotonically to $f \in \mathcal{B}$.

Proof. We use Theorem 2.6 to show the existence of an extension. In the notation of Theorem 2.6, for any $f \in \mathcal{B}$ we define $f(L)$ by

$$U f(L) U^{-1} \phi = (f \circ h) \phi. \quad (2.4)$$

It is straightforward to check that it satisfies properties 1-5 in Theorem 2.4. And the limiting behaviour in 6 is the consequence of dominated convergence theorem.

Now we show the uniqueness. Note that the uniqueness does not follow without addition requirement 6. Given two maps $f \mapsto f(L)$ with properties 1-6, let $\mathcal{C}$ be the set on which the two map coincides. It is clear that $C_0(\mathbb{R}) \subset \mathcal{C}$. by part 6, $\mathcal{C}$ should be closed under monotone limits of sequences in $C_0(\mathbb{R})$. But it gives the whole space $\mathcal{B}$. $\square$

Example 2.1. Let L be a self-adjoint operator on $\mathcal{H}$. For any given $t \in \mathbb{R}$, $e^{itx} \in \mathcal{B}$. Hence we can define e^{iLt} and show that $f(t) := e^{iLt} f$ is a solution to the Schrödinger equation $\frac{df}{dt} = iLf$. See Exercise 2.15 for details.

Definition 2.4: Unitary Group

A **one-parameter unitary group** is a map $t \mapsto U_t$ from $\mathbb{R}$ to $\mathcal{L}(\mathcal{H})$ such that

$$U_0 = 1, \quad U_{t+s} = U_t U_s, \quad U_t^* = U_{-t},$$

and $t \mapsto U_t$ is continuous in the strong topology of operators, i.e. $U_t \xrightarrow{s} U_{t_0}$ when $t \rightarrow t_0$.

It is obvious from the definition that U_t for all $t \in \mathbb{R}$ is a unitary operator.

$$U_t U_t^* = U_t U_{-t} = 1 = U_t^* U_t.$$

Remark 2.7. For any self-adjoint operator L , the one parameter family $\{e^{iLt}\}$ defines a unitary group. See Exercise 2.15.

Let's calculate the unitary group for some of the examples we have seen.

Example 2.2. a) *Position Operator.* Consider the self-adjoint operator $L = M_x$ on $L^2(\mathbb{R})$ with $D(M_x) = \{\phi \in L^2(\mathbb{R}) : x\phi \in L^2(\mathbb{R})\}$. Then the associated unitary group is given by

$$U_t f(x) = e^{itM_x} f(x) = e^{ixt} f(x) = M_{e^{ixt}} f(x), \quad \forall f \in L^2(\mathbb{R}).$$

Note that the last but one identity follows from (2.4).

b) *Momentum Operator.* Consider the self-adjoint operator $L = -i\frac{d}{dx}$ on $L^2(\mathbb{R})$ with $D(L) = \{f \in L^2(\mathbb{R}) : \hat{\xi}f \in L^2(\mathbb{R})\}$. Recall that L is unitary equivalent to M_ξ :

$$\mathcal{F}L\mathcal{F}^{-1}\hat{f} = \xi\hat{f}.$$

Then the associated unitary group is given by $\mathcal{F}U_t\mathcal{F}^{-1}\hat{f} = M_{e^{it\xi}}\hat{f}$. Hence,

$$U_t f(x) = \mathcal{F}^{-1}(e^{it\xi}\hat{f})(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{i(x+t)\xi} \hat{f}(\xi) d\xi = f(x+t).$$

It shows that U_t is simply a translation operator mapping $f(x)$ to $f(x+t)$.

c) *Laplacian.* Consider Δ on $L^2(\mathbb{R}^n)$ with domain $H^2(\mathbb{R}^n)$. Recall from Example 1.7 and Exercise 1.1 that

$$\mathcal{F}\Delta\mathcal{F}^{-1}\hat{f} = |\xi|^2\hat{f}, \quad \hat{f} \in \mathcal{D}(M_{|\xi|^2}).$$

Hence,

$$e^{it\Delta}f(x) = \mathcal{F}^{-1}(e^{it|\xi|^2}\hat{f}).$$

For $\hat{f}, \hat{g} \in L^2(\mathbb{R}^n)$, one can show that

$$\mathcal{F}^{-1}(\hat{f}\hat{g}) = \frac{1}{(2\pi)^{n/2}} f * g.$$

However, $e^{it|\xi|^2} \notin L^2(\mathbb{R})$. Let us consider $g_k(\xi) = e^{(-\frac{1}{k}+it)|\xi|^2}$. Then $g_k \in L^2(\mathbb{R})$ for every $k \in \mathbb{N}$. Since $e^{(-\frac{1}{k}+it)x}$ converges monotonically to e^{itx} ,

$$e^{(-1/k+it)\Delta} \xrightarrow{s} e^{it\Delta}.$$

So we first calculate $e^{(-1/k+it)\Delta}$ and then pass to the limit. Similar to the above, we have

$$e^{(-1/k+it)\Delta}f(x) = \mathcal{F}^{-1}(e^{(-1/k+it)|\xi|^2}\hat{f}).$$

$\mathcal{F}^{-1}(g_k)(x) = \frac{1}{(2z_k)^{n/2}} e^{\frac{|x|^2}{4z_k}}$, where $z_k = -1/k + it$. Therefore,

$$e^{(-1/k+it)\Delta}f(x) = \frac{1}{(4\pi z_k)^{n/2}} \int_{\mathbb{R}^n} e^{\frac{|x-y|^2}{4z_k}} f(y) dy \rightarrow \frac{1}{(4\pi it)^{n/2}} \int_{\mathbb{R}^n} e^{\frac{|x-y|^2}{4it}} f(y) dy = e^{it\Delta}f(x).$$

Here $i^{1/2} = e^{i\pi/4}$.

The following theorem is very important in quantum mechanics. You may view it as a reverse to Remark 2.7.

Theorem 2.8: Stone's Theorem

Let $\{U_t\}_{t \in \mathbb{R}}$ be a one-parameter unitary group. Then there is a self-adjoint operator L on $\mathcal{H}$ such that $U_t = e^{itL}$ for all $t \in \mathbb{R}$. Moreover,

$$Lf = \lim_{t \rightarrow 0} \frac{(U_t - 1)f}{it}$$

with

$$\mathcal{D}(L) := \left\{ f \in \mathcal{H} : \lim_{t \rightarrow 0} \frac{(U_t - 1)f}{it} \text{ exists} \right\}.$$

The operator L is called the **generator** of $\{U_t\}$. Stone's theorem shows how an unbounded self-adjoint operator naturally appears as the generator of a unitary group.

2.3 Spectral Projection

The last version of the spectral theorem we state here is in parallel to third version of Spectral theorem for Hermitian matrices. We recall the definition of orthogonal projection.

Definition 2.5: Spectral Projection

An **orthogonal projection** $P : \mathcal{H} \rightarrow \mathcal{H}$ is a bounded linear operator such that

$$P = P^* = P^2.$$

Let L be a self-adjoint operator, $A \subset \mathbb{R}$ be any Borel set and χ_A its characteristic function. We can define $P_A(L) = \chi_A(L)$. It is an orthogonal projection since $\chi_A = \chi_A^2 = \bar{\chi}_A$. $\{P_A(L)\}$ are called the **spectral projections** for L .

Spectral projections give rise to a projection-valued measure. Part 6 of Theorem 2.7 implies for Borel sets $\{A_j\}_{j=1}^\infty$ with $A_j \cap A_k = \emptyset$ when $j \neq k$ we have

$$\sum_{j=1}^N P_{A_j}(L) \xrightarrow{s} P_{\cup_{j=1}^\infty A_j}(L)$$

The map $A \mapsto P_A(L)$ is called a **projection-valued measure**.

Let us define $E_\lambda := P_{(-\infty, \lambda]}(L)$. The family $\{E_\lambda\}$ gives a so-called **resolution of identity** associated to L . Moreover, we can write

$$\langle Lf, g \rangle = \int_{\mathbb{R}} \lambda \langle dE_\lambda f, g \rangle;$$

$$L = \int_{\mathbb{R}} \lambda dE_\lambda;$$

and

$$f(L) = \int_{\mathbb{R}} f(\lambda) dE_\lambda, \quad f \in C_0(\mathbb{R}).$$

You may compare the above identities with the first and third form of the spectral theorem for Hermitian matrices discussed in the beginning of this chapter. For the discussion on the

definition of a resolution of identity and how we view the above integrals, see [Hel13, Section 8.3] and [Sim15, Section 5.1].

We can also use the spectral projections to define an important spectral decomposition. We have already named to an important subset of $\sigma(L)$: the point spectrum $\sigma_p(L)$ which is the set of eigenvalues of a self-adjoint operator L .

Definition 2.6: Discrete and Essential Spectrum

For a self-adjoint operator L , the **discrete spectrum** $\sigma_{\text{disc}}(L)$ is the set of all eigenvalues λ with finite multiplicities that are isolated, i.e. there exists $\epsilon > 0$ such that

$$(\lambda - \epsilon, \lambda + \epsilon) \setminus \{\lambda\} \cap \sigma(L) = \emptyset.$$

The complement of the discrete spectrum is called the **essential spectrum** and we denote it by $\sigma_{\text{ess}}(L)$.

Another way to define $\sigma_{\text{disc}}(L)$ and $\sigma_{\text{ess}}(L)$ is by the spectral projections.

Lemma 2.4. The essential spectrum $\sigma_{\text{ess}}(L)$ is the set of $\lambda \in \sigma(L)$ such that $P_{(\lambda-\epsilon, \lambda+\epsilon)}$ has infinite rank for all $\epsilon > 0$. The discrete spectrum $\sigma_{\text{disc}}(L)$ is the set of $\lambda \in \sigma(L)$ such that $P_{(\lambda-\epsilon, \lambda+\epsilon)}$ has finite rank for some $\epsilon > 0$.

The proof of this lemma is a consequence of the following useful lemma in measure theory and Theorem 2.6.

Lemma 2.5. Let μ be a finite measure on $\mathbb{R}^n$ and X a Borel subset of $\mathbb{R}^n$. Then $L^2(X, d\mu)$ has finite dimension if and only if there exists distinct points $\{x_1, \dots, x_k\} \subset X$ such that $X \setminus \{x_1, \dots, x_k\}$ has zero measure.

In Theorem 1.4, we showed that elements in the spectrum can be seen as approximate eigenvalues. If we add an extra assumption to that theorem we can characterise the essential spectrum in the following way.

Theorem 2.9: Weyl's Criterion

Let L be a self-adjoint on $\mathcal{H}$. A point $\lambda \in \sigma_{\text{ess}}(L)$ if and only if there exists a sequence (x_n) in $\mathcal{D}(L)$ with $\|x_n\| = 1$, $x_n \rightarrow 0$ in the weak sense and $\|(L - \lambda)x_n\| \rightarrow 0$.

We see that the extra assumption here is the weak convergence of x_n to 0.

Proof. If $\lambda \in \sigma_{\text{ess}}(L)$ then the range of $P_{(\lambda-1/n, \lambda+1/n)}$, $n \in \mathbb{N}$, is infinite dimensional. For each $n \in \mathbb{N}$, take $x_n \in \text{Ran}(P_{(\lambda-1/n, \lambda+1/n)})$ such that it is orthogonal to x_j , $j \in \{1, \dots, n-1\}$. It is easy to show that x_n weakly converges to 0. Since $x_n \in \text{Ran}(P_{(\lambda-1/n, \lambda+1/n)})$, we have

$$\|(L - \lambda)x_n\| < 1/n.$$

If $\lambda \in \sigma_{\text{disc}}(L)$, then $\text{Ran}(P_{(\lambda-\epsilon, \lambda+\epsilon)})$ has finite rank for some $\epsilon > 0$. Let $v_1, \dots, v_k$ be a basis for $\text{Ran}(P_{(\lambda-\epsilon, \lambda+\epsilon)})$. Take any sequence (x_n) in $\mathcal{D}(L)$ such that $\|x_n\| = 1$ and $x_n \rightarrow 0$ weakly. Then

$$\|P_{(\lambda-\epsilon, \lambda+\epsilon)}x_n\| \rightarrow 0$$

since $\langle x_n, v_j \rangle \rightarrow 0$ for all $1 \leq j \leq k$. Note that $P_A(L)$ commutes with L .

$$\begin{aligned} \|(L - \lambda)x_n\| &= \|(L - \lambda)(1 - P_{(\lambda-\epsilon, \lambda+\epsilon)})x_n\| + \|(L - \lambda)P_{(\lambda-\epsilon, \lambda+\epsilon)}x_n\| \\ &\geq \epsilon\|(1 - P_{(\lambda-\epsilon, \lambda+\epsilon)})x_n\| - \epsilon\|P_{(\lambda-\epsilon, \lambda+\epsilon)}x_n\| \\ &\geq \epsilon. \end{aligned}$$

□

For eigenvalues of positive self-adjoint operators with compact resolvent, the Min-Max Principle gives a variational characterisation. The following theorem is an extension of the min-max theorem we have seen earlier to all positive self-adjoint operators. It gives a characterisation for the bottom of the essential spectrum which is of great interest both physically and geometrically.

Theorem 2.10: Min-Max Principle II

Let L be a non-negative self-adjoint operator on $\mathcal{H}$. Consider the sequence of λ_k defined in (2.1). If $\sigma_{\text{ess}}(L) \neq \emptyset$, then one of the following situations happens.

- i) There exists $c \in \mathbb{R}$ such that $\lambda_k < c$ for all $k \in \mathbb{N}$ and $\lim_{k \rightarrow \infty} \lambda_k = c$. Then $\sigma(L) \cap [0, c) = \{\lambda_k\}_{k=1}^{\infty} \subset \sigma_{\text{disc}}(L)$ and $c = \min\{t : t \in \sigma_{\text{ess}}(L)\}$.
- ii) There exists $c \in \mathbb{R}$ and $N \in \mathbb{N}$ such that $\lambda_N < c$ but $\lambda_k = c$ for all $k > N$. Then $\sigma(L) \cap [0, c) = \{\lambda_1, \dots, \lambda_N\} \subset \sigma_{\text{disc}}(L)$ and $c = \min\{t : t \in \sigma_{\text{ess}}(L)\}$.

We end this section with the following nice theorem which show the relation between the spectrum of L and $f(L)$.

Theorem 2.11: Spectral Mapping Theorem

Let L be a self-adjoint operator and f be a bounded continuous function on $\sigma(L)$ if L is bounded, and on $\sigma(L) \cup \{\infty\}$ if L is unbounded. Then if L is bounded

$$\sigma(f(L)) = f(\sigma(L));$$

and if L is unbounded

$$\sigma(f(L)) = f(\sigma(L) \cup \{\infty\}).$$

Proof. We first prove it when L is bounded. Let $z \notin f(\sigma(L))$. Note that $\sigma(L)$ is compact. Hence $\text{dist}(z, f(\sigma(L))) > 0$. Then $M_{(z-f(x))^{-1}}$ is bounded operator on each $L^2(\sigma(L), d\mu_n)$, where μ_n is the restriction of μ to $\sigma(L) \times \{n\}$, and

$$\|M_{(z-f(x))^{-1}}\| \leq \frac{1}{\text{dist}(z, f(\sigma(L)))}.$$

In the notations of Sections 2.2.2 & 2.2.3, we conclude that $U^{-1}M_{(z-f(x))^{-1}}U$ is the inverse of $z - f(L)$ for $z \notin f(\sigma(L))$. Therefore, $z \notin \sigma(f(L))$.

Let $\lambda = f(t_0) \in f(\sigma(L))$, where $t_0 \in \sigma(L)$. f is continuous, therefore for every $\epsilon > 0$, there exists $\delta > 0$ such that for all $|t - t_0| < \delta$

$$|f(t) - \lambda| < \epsilon.$$

Since $P_{(t_0-\delta, t_0+\delta)} \neq 0$, we can take a unit vector $v \in \text{Ran}(P_{(t_0-\delta, t_0+\delta)})$. We have

$$\|(f(L) - \lambda)v\| \leq \epsilon\|v\| = \epsilon.$$

Therefore, $\lambda \in \sigma(f(L))$.

If L is unbounded the proof remain the same. We use that fact that $\sigma(L) \cup \{\infty\}$ is compact and we can repeat the argument. $\square$

2.4 Exercises

2.1 The goal is to prove Theorem 2.1. Under the assumption of Theorem 2.1

- a) show that $\|L\|$ or $-\|L\|$ is an eigenvalue.
- b) Show that for every $c > 0$ there is only finitely many eigenvalues λ_n such that $|\lambda_n| > c$.
- c) Let e_1 be the unit vector correspond to the eigenvalue $\|L\|$ or $-\|L\|$. Show that $\text{span}(e_1)^\perp$ is invariant under L .
- d) Produce an orthonormal set $\{e_1, \dots, e_k\}$ of eigenvectors with eigenvalues

$$\|L\| = |\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|.$$

- e) Conclude that $L_k(x) := \sum_{n=1}^k \lambda_n \langle x, e_n \rangle e_n$ converges in norm to L .

2.2 a) Let L be a compact operator. Show that $0 \in \sigma(L)$.

b) Give an example of a compact operator such that $\sigma_p(L) = \emptyset$.

2.3 Assume that the operator L has a compact resolvent, i.e. $(z_0 - L)^{-1}$ is compact for some $z_0 \in \rho(L)$. Show that for every $z \in \rho(L)$, $(z - L)^{-1}$ is compact.

2.4 Prove Corollary 2.1.

Hint: You may use the fact that T is compact if and only if it can be written as the limit of bounded operators with finite rank.

2.5 Show that the inclusion $H_0^1(\mathbb{R}) \hookrightarrow L^2(\mathbb{R})$ is not compact.

2.6 Prove Theorem 2.3.

2.7 Let Ω be a bounded domain in $\mathbb{R}^n$ with Lipschitz boundary. Show that

$$\lambda_k(\Delta_N) \leq \lambda_k(\Delta_D)$$

where $\lambda_k(\Delta_D)$ and $\lambda_k(\Delta_N)$ are the k -th eigenvalue of the Dirichlet Laplacian and Neumann Laplacian of Ω respectively.

Hint. You may use Theorem 2.3 and Remark 2.3.

2.8 Show that $|\frac{\partial \tilde{f}(z)}{\partial \bar{z}}| = O(|y|^n)$ as $y \rightarrow 0$. In particular $\frac{\partial \tilde{f}(z)}{\partial \bar{z}} = 0$ for all $z \in \mathbb{R}$.

2.9 Let (X, μ) be a σ -finite measure space and $\phi : X \rightarrow \mathbb{R}$ be a real-valued measurable function. Consider the multiplication operator M_ϕ on $L^2(X, \mu)$ with $\mathcal{D}(M_f) = \{f \in L^2(X, \mu) : \phi f \in L^2(X, \mu)\}$.

a) Show that M_ϕ is self-adjoint.

b) Show that

$$\sigma(M_\phi) = \{\lambda \in \mathbb{R} : \mu(\phi^{-1}(\lambda - \epsilon, \lambda + \epsilon)) > 0, \forall \epsilon > 0\} =: \text{Ran}_{\text{ess}}(\phi).$$

This set is called the essential range of ϕ .

c) Show that for $z \in \rho(M_\phi)$

$$\|(z - M_\phi)^{-1}\| = \frac{1}{\text{dist}(z, \sigma(M_\phi))}.$$

Hint: The proof of parts a)-b) is similar to Example 1.5 and Exercise 1.1.

2.10 Let L be a self-adjoint operator. Show that for every $f \in C_0(\mathbb{R})$,

$$\|f(L)\| = \sup_{x \in \sigma(L)} |f(x)|.$$

2.11 Show that in Definition 2.2 when L_n and L are bounded self-adjoint operators then the convergence in resolvent norm is equivalent to the convergence in operator norm.

2.12 Let L_n and L be positive self-adjoint operators. Then L_n converges to L in the norm resolvent sense if and only if

$$\|(L_n + 1)^{-1} - (L + 1)^{-1}\| \rightarrow 0.$$

2.13 Prove the statement given in Remark 2.6.

2.14 Prove Lemma 2.3.

2.15 Let L be a self-adjoint operator on $\mathcal{H}$. For any given $t \in \mathbb{R}$, consider the operator e^{iLt} .

a) Show that e^{iLt} is a unitary operator.

b) Prove that $t \rightarrow e^{iLt}$ is a unitary group, i.e.

$$e^{iL(t+s)} = e^{iLt}e^{iLs}, \quad \forall t, s \in \mathbb{R}.$$

c) Prove that $\lim_{t \rightarrow s} \|e^{iLt}f - e^{iLs}f\| = 0$ for all $s \in \mathbb{R}$ and all $f \in \mathcal{H}$.

d) Prove that if $f \in \mathcal{D}(L)$, then $f(t) := e^{iLt}f$ is a solution of the Schrödinger equation

$$\frac{df}{dt} = iLf.$$

2.16 Prove Corollary 2.2.

Hint: You may use Theorem 2.4 and apply it to $f \in C_c^\infty(\mathbb{R})$ with $f(\lambda) = 1$. You need to carefully choose the support of f .

2.17 Let P an orthogonal projection on $\mathcal{H}$. Identify $\sigma_{\text{ess}}(P)$ and $\sigma_{\text{disc}}(P)$.

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